AIRWise: Smart Room-Responsive Remote-Control System for Thermal Comfort

---

A Design / Capstone Project

Presented to the Department of Electronics Engineering

Cebu Institute of Technology University

Cebu City, Philippines

In Partial Fulfillment

of the Requirements for the Degree

Bachelor of Science in Electronics Engineering

---

by

STFU

Hyuan Krisjae A. Telmoso

Daryl Jay D. Gimperoso

Lloyd Sedrick M. Salandron

January 2026

Approval Sheet

This thesis entitled, “TITLE OF THE STUDY”, prepared and submitted by Complete Name of Member A, Complete Name of Member B, Complete Name of Member C, and Complete Name of Member D in partial fulfillment of the requirements for the degree BACHELOR OF SCIENCE IN ELECTRONICS ENGINEERING is hereby recommended for approval.

Dr. Rachel M. Chong, ECE
Adviser
Date: _____

Engr. Panelist A Complete Name
Member
Date: _____

Engr. Panelist B Complete Name
Member
Date: _____

Engr. Panelist C Complete Name
Member
Date: _____

This thesis is approved in partial fulfillment of the requirements for the degree of BACHELOR OF SCIENCE IN ELECTRONICS ENGINEERING.

Dr. Rachel M. Chong, ECE
Instructor
Date: _____

Engr. Susana A. Tan, PECE, MSManE
Chair, Department of Electronics Engineering
Date: _____

Dr. Evangeline C. Evangelista
Dean, College of Engineering and Architecture
Date: _____

Abstract

This is a sample file for the abstract of your thesis. This is a chapter with no number. Create your abstract AFTER ALL CHAPTERS are already completed.

Your abstract should contain a maximum of 500 words. All texts must be plain. This means that words must not be bold or italicized. Citations should also be avoided here.

This chapter will be required when the project is already complete. The paragraphs below will be your guide on the contents of your abstract

An abstract is a short summary of your completed research. If done well, it makes the reader want to learn more about your research.

These are the basic components of an abstract in any discipline:

1. Motivation/problem statement:

   Why do we care about the problem? What practical, scientific, theoretical or artistic gap is your research filling?

2. Methods/procedure/approach:

   What did you actually do to get your results? (e.g. analyzed 3 novels, completed a series of 5 oil paintings, interviewed 17 students)

3. Results/findings/product:

   As a result of completing the above procedure, what did you learn/invent/create?

4. Conclusion/implications:

   What are the larger implications of your findings, especially for the problem/gap identified in the motivation/problem statement?

However, it is important to note that the weight accorded to the different components can vary by discipline. For models, try to find abstracts that is similar to

your research or project.

keywords: Select specific words or group of words that are accurately describes this thesis. Up to ten (10) keywords can be enumerated, separated by a semi-colon (;).

Acknowledgment

This will contain the acknowledgments. This chapter has also no number and all texts must be plain.. In this chapter, all persons who helped towards the success of your thesis should be thanked for their contributions. Each member can have their own paragraph but this must be edited in order to have a continuity of thoughts. Grants and special funding that you have received to support the research should also be mentioned here.

This can contain several paragraphs and can span up to two pages. This chapter will be required in PROJECT 2, i.e. when the project is already complete.

Table of Contents

		Page
Abstract		**iii**
Acknowledgment		**v**
List of Tables		**ix**
List of Figures		**xi**
List of Figures		**xiii**
Chapter 1	**INTRODUCTION**	**1**
1.1	Background of the Study	1
1.2	Problem Statement	2
1.3	Objectives	2
1.4	Purpose of the Study	3
1.4.1	Sustainable Development Goals	3
1.4.2	Building Managers and Maintenance	3
1.4.3	Aircon Users	3
1.4.4	Technology Developers	4
1.4.5	Future Researchers	4
1.5	Definition of Terms	4
Chapter 2	**THEORETICAL BACKGROUND**	**5**
2.1	Principles of Control Systems in HVAC	5
2.1.1	Feedback Control Theory	5
2.1.2	Proportional-Integral-Derivative (PID) Control	6
2.1.3	Occupancy-Responsive Logic and Automation	6
2.2	Infrared (IR) Communication Protocol	7

2.2.1 Signal Modulation and Carrier Frequency 7
2.2.2 Pulse Distance Width Modulation 8
2.2.3 Data Frame Structure and Checksums 8
2.2.4 Hardware Interfacing: The Optoelectronic Link 9
2.2.5 Operational Mode Configurations in Air Conditioning Systems 9
2.2.6 Dehumidification and Airflow Dynamics 10
2.2.7 Kinematic of Swing Mode 11
2.2.8 Sensors 12
2.2.9 Human Thermal Comfort 15
2.2.10 Regional Climate Constraints and Thermal Sensation in the Philippines 17
2.2.11 ANSI/ASHRAE Standard 55 Framework 18
2.2.12 ANSI/ASHRAE Standard 55 Framework 18
2.2.13 Predicted Percentage Dissatisfied (PPD) Index 19
2.3 Literature Review 19
2.3.1 Manual Remote-Control System 19
2.3.2 Automatic Remote-Control Systems 20
Chapter 3 RESEARCH METHODOLOGY 23
3.1 Proposed Method 23
3.1.1 System Components 23
3.1.2 Prototype 29
3.1.3 Algorithm 31
3.1.4 Circuit Configuration 37
3.2 Description of Experiments 40
3.2.1 Temperature and Humidity Sensor Accuracy Test 40
3.2.2 Motion Sensor Accuracy Test 46

3.2.3 Wireless Sensor Node Performance and Data Accuracy Test 52
3.2.4 Evaluation of AIRWise IR transmission Performance to Panasonic Window-type Inverter Air conditioner 55
3.2.5 Voting Mechanism Accuracy Test 56
Chapter 4 RESULTS AND DISCUSSIONS 59
4.1 Temperature and Humidity Sensor Accuracy Test 59
4.2 Motion Sensor Accuracy Test 61
4.3 Wireless Sensor Nodes Transmission Test 62
Chapter 5 CONCLUSIONS AND RECOMMENDATIONS 64
5.1 Conclusions 64
5.2 Recommendations 64
BIBLIOGRAPHY 65
Appendix A PRIOR ART SEARCH and ANALYSIS REPORT 69
Appendix A PATENT DRAFT 76
Appendix B TIMETABLE OF ACTIVITIES 88
Appendix C RESEARCH BUDGET 91
Appendix D ADDITIONAL FLOWCHARTS 92
Appendix E PSEUDO-CODES 93
Appendix F ELECTRICAL CIRCUITS 94
Appendix G SUPPLEMENTAL DATA 95
Appendix H PLAGIARISM CHECK REPORT 96
Appendix I ETHICS CLEARANCE 104
Appendix J CURRICULUM VITAE 106

List of Tables

Page

3.1 DHT22 Sensor Accuracy at x°C 45

3.2 DHT22 Sensor Accuracy at x% 45

3.3 V outputs 1 meter at 0° 48

3.19 V PIR Motion detection results at 1 m distance 48

3.4 V outputs 1 meter at $\pm 15^{\circ}$ 48

3.5 V outputs 1 meter at $\pm 30^{\circ}$ 49

3.6 V outputs 1 meter at $\pm 45^{\circ}$ 49

3.7 V outputs 2 meters at 0° 49

3.8 V outputs 2 meters at $\pm 15^{\circ}$ 49

3.9 V outputs 2 meters at $\pm 30^{\circ}$ 49

3.10 V outputs 2 meters at $\pm 45^{\circ}$ 49

3.11 V outputs 3 meters at 0° 50

3.12 V outputs 3 meters at $\pm 15^{\circ}$ 50

3.13 V outputs 3 meters at $\pm 30^{\circ}$ 50

3.14 V outputs 3 meters at $\pm 45^{\circ}$ 50

3.15 V outputs 4 meters at 0° 50

3.16 V outputs 4 meters at $\pm 15^{\circ}$ 50

3.17 V outputs 4 meters at $\pm 30^{\circ}$ 51

3.18 V outputs 4 meters at $\pm 45^{\circ}$ 51

3.20 V PIR Motion detection results at 2 m distance 51

3.21 V PIR Motion detection results at 3 m distance 51

3.22 V PIR Motion detection results at 4 m distance 52

3.23 Sensor Node 1 packet validation 54
3.24 Sensor Node 2 packet Validation 54
3.25 Sensor Node 3 Packet Validation 54
3.26 Transmission Success Rate per Sensor Node 54
3.27 AIRWise evaluation test 56
3.28 Voting mechanism accuracy test 58

4.1 Motion Performance at 1-meter Distance 62
4.2 Motion Performance at 2-meter Distance 62
4.3 Motion Performance at 3-meter Distance 62
4.4 Motion Performance at 4-meter Distance 63

List of Figures

Page

2.1 NEC Protocol at 67.5 ms [4] 8

2.2 Vapor Compression Refrigeration Cycle [5] 10

3.1 Simplified Block Diagram of the System 24

3.2 DHT22 temperature and humidity sensor 25

3.3 HCSR501 PIR motion sensor 25

3.4 Self-reset switch 26

3.5 940 nm IR LED 26

3.6 1.3” OLED Display I2C 27

3.7 ESP32 Microcontroller 28

3.8 ESP8266 Node MCU 29

3.9 Panasonic CW-XU1021VPH window-type air conditioner 29

3.10 Sensor Node 30

3.11 AIRWise Control Hub 30

3.12 Sensor Nodes 1,2,3 Flow 32

3.13 CRC_{rec} Subprocess 33

3.14 Control Hub Flow 34

3.15 Swing control subroutine 36

3.16 Error-handling subroutine 37

3.17 Voting mechanism flow 38

3.18 Sensor Nodes 1,2,3 Schematic Diagram 39

3.19 Control Hub Full Schematic Diagram 41

3.20 IR receiver schematic Diagram 42

4.1 DHT22 (Temperature) Linear curve 60

4.2 DHT22 (Humidity) Linear curve 61

List of Symbols

$\%A$	percent accuracy
α	Seebeck coefficient
ΔT	temperature change (°C)
η	responsivity of the photodiode, amperes per watt (A/W)
Off	system turned off state
θ	angle from sensor centerline
ε_0	permittivity of free space
ε_r	dielectric constant
A	expected value
A	surface area of electrodes
B	actual value
C_p	specific heat capacity of air, joules per kilogram-kelvin (J/kg·K)
d	distance between electrodes
d	distance from the sensor
f_c	carrier frequency, Hertz
k	material constant
K_d	Derivative gain
K_i	Integral gain

K_p Proportional gain

m mass flow rate of air, kilograms per second (kg/s)

N rotational speed

N_{high} number of trials with output voltage ≥ 2.5 V

N_{scs} number of packets successfully received by the control hub

P shaft power

$P(T)$ saturation vapor pressure at temperature T

P_{in} incident optical power, watts (W)

Ph partial pressure of water vapor

Q heat transfer rate, watts (W) or BTU/hr

R_0 base resistance

R_f feedback resistance, ohms (Ω)

S occupancy status

T_b baseline temperature (°C)

T_c carrier period, microseconds

$T_{comfort}$ comfort temperature

$T_{standby}$ standby temperature

T_{cold} cold junction temperature, Kelvin

T_{hot} hot junction temperature, Kelvin (K)

T_{in} inlet air temperature, Kelvin (K)

T_{out} outlet air temperature, Kelvin (K)

TSR transmission success rate (%)

Chapter 1

INTRODUCTION

This chapter provides an overview of current trends in air conditioning systems and indoor climate management, focusing on common usage patterns, existing inefficiencies, and emerging solutions. It outlines the background and motivation for the study, identifies the specific challenges addressed by this research, and reviews relevant technological advances. The chapter presents the core objectives that will guide the study's objectives.

1.1 Background of the Study

The use of air conditioning in homes and offices has made indoor life much healthier and more productive. However, using these systems the wrong way leads to high power bills and environmental issues [7]. Research shows that common mistakes like setting the temperature too low, constantly turning the unit on and off, or ignoring energy-saving settings-greatly increase electricity use [8].

Ultimately, how people use their aircon is often more important than the machine's technology. Even the most advanced units can't save energy if they aren't used correctly. Many people ignore features like adjustable fan speeds or swing settings, missing out on ways to stay cool more efficiently [19]. Over time, these bad habits do more than just waste money; they also lead to poorer air quality and cause the equipment to wear out faster.

A major problem is that people often leave the air conditioning on in empty rooms [9]. Studies on energy use show that up to 50% of cooling energy is wasted on rooms where no one is present. This happens because people often forget to turn units off, lack awareness about the impact, or simply don't have easy tools to

automate their AC [10].

Because the global demand for cooling is growing, developing remote controls that respond to whether a room is occupied is one of the most important goals for modern indoor engineering.

1.2 Problem Statement

A significant limitation of manually operated air-conditioning units is their static delivery of cooling, which fails to account for shifting thermal loads or occupancy patterns. Without adaptive control, these systems often run a higher intensity that required, leading to substantial energy waste. Furthermore, fixed manual settings cannot compensate for internal heat gains, fluctuating outdoor temperatures, or individual user preferences. This lack of responsiveness creates a dual failure: it inflates operational costs through energy inefficiency and diminishes indoor environmental quality through inconsistent temperature control.

1.3 Objectives

The objective of the study is to design a smart, room-responsive remote-control system for air conditioning unit that automatically adjusts mode settings, fan speed, swing, and temperature based on room's ambient conditions. The specific objectives are as follows:

- create a network that collect and monitor room ambient conditions using a set of wireless sensor nodes for data transmission;
- develop an intelligent decision system capable of determining and adjusting optimal air conditioning settings based on data collected from the wireless sensor nodes;

- design a wireless control hub that transmit the computed air conditioning control settings from the intelligent decision system to the aircon.

1.4 Purpose of the Study

The purpose of this study will provide the benefits of the following:

1.4.1 Sustainable Development Goals

This study contributes to three key United Nations Sustainable Development Goals by advancing intelligent air conditioning systems for buildings. It promotes SDG 7 by enabling more efficient use of energy, supports SDG 9 through the integration of innovative control technologies within modern infrastructure, and advances SDG 12 by encouraging responsible consumption and minimizing energy waste. Through these efforts, the research addresses affordability, innovation, and sustainability in building climate management.

1.4.2 Building Managers and Maintenance

The proposed system reduces maintenance burdens and operational expenses by enabling automated optimization and early fault detection, making building management simpler and more cost-effective.

1.4.3 Aircon Users

Aircon users benefit from consistent thermal comfort and lower electricity bills, as the study's smart technology effortlessly delivers optimal cooling tailored to real-time needs.

1.4.4 Technology Developers

Developers gain new opportunities to design, implement, and improve intelligent control hardware and software for climate management in future smart buildings.

1.4.5 Future Researchers

The study provides foundational knowledge and tested methods that future researchers can build upon, facilitating further innovation in energy-efficient building controls and sustainable environmental technologies.

1.5 Definition of Terms

In the context of this project, the following are the operational definitions of terms:

AIRWise: an automated intelligent remote control for a wise indoor environment.

AC (Aircon): a system for cooling indoor places

Sensor Node: a wireless device that detects motion, humidity, and temperature in the system

Chapter 2

THEORETICAL BACKGROUND

This chapter explores into a comprehensive review of theories and literature, including principles, concepts, and ideas applicable in the study. The first part of this chapter includes the discussion of the concept of control systems, wireless communications, Mode and settings of an air conditioning unit, working principle of sensors and system outputs. The second part of this chapter includes the review of related literature.

2.1 Principles of Control Systems in HVAC

The fundamental objective of a control system in Heating, Ventilation, and Air Conditioning (HVAC) is to maintain a process variable – typically temperature or humidity – at desired setpoint by manipulating the system's output [1]. In conventional manual systems, this control loop is often or inefficiently closed, leading to performance gaps.

2.1.1 Feedback Control Theory

A closed-loop control system utilizes feedback to reduce the error between the measured variable (y) and the reference setpoint (r). The controller calculates an error signal, $e(t) = r(t) - y(t)$, and adjusts the actuator accordingly [2].

In air conditioning applications, the "plant" represents the indoor environment, while the "actuator" is the compressor or fan motor. Traditional feedback systems are limited because they assume a constant state, failing to adjust when the thermal load drops to zero in unoccupied spaces [11].

2.1.2 Proportional-Integral-Derivative (PID) Control

For precise climate management, PID control is the industry standard. This algorithm calculates the control output $u(t)$ based on three distinct terms:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d \frac{de(t)}{dt} \tag{2.1}$$

where:

K_p (Proportional) = adjust the output based on the current magnitude of the error;

K_i (Integral) = eliminates steady-state error by accounting for past errors; and

K_d (Derivative) = predicts future error by analyzing the rate of change [20].

While PID controllers offer stable temperature regulation, they are inherently 'reactive.' PID controller alone cannot address energy waste if the setpoint $r(t)$ remains active in an empty room.

2.1.3 Occupancy-Responsive Logic and Automation

The integration of occupancy data into the control loop transforms a standard feedback system into a "context-aware" system. This is often achieved through State-Space-Control or Rule-Based-Logic [12]. In these systems, the occupancy sensor acts as binary switch (S) or a continuous variable that modifies the controller's state:

$$Set\ Point = \begin{cases} T_{comfort}, & \text{if } S = 1 \text{ (Occupied)} \\ T_{standby}, Off, & \text{if } S = 0 \text{ (Unoccupied)} \end{cases} \tag{2.2}$$

where:

$T_{comfort}$ = comfort temperature when occupied;

$T_{standby}$ = standby temperature when unoccupied;

Off = system turned off state; and

S = occupancy status.

According to recent studies, implementing such occupancy-responsive thresholds can reduce HVAC energy consumption by 20% to 50% depending on the building type [13].

2.2 Infrared (IR) Communication Protocol

Infrared (IR) communication is a wireless technology utilizing light waves in the electromagnetic spectrum, typically between 800 nm and 1000 nm [3]. In AC systems, IR serves as the physical layer interface for command transmission.

2.2.1 Signal Modulation and Carrier Frequency

To ensure signal integrity against ambient infrared noise (e.g., sunlight or fluorescent lightning), the data signal must be modulated onto a carrier frequency, typically fc = 38 kHz. The period of this carrier signal is defined as:

$$T_c = \frac{1}{f_c} \approx 26.3\ \mu s \tag{2.3}$$

where:

T_c = carrier period, microseconds and

f_c = carrier frequency, Hertz.

The modulation process involves turning the IR LED on and off at this frequency. A logical ‘1’ or ‘0’ is not presented by a simple voltage level, but by the duration of these bursts or the space between them [4].

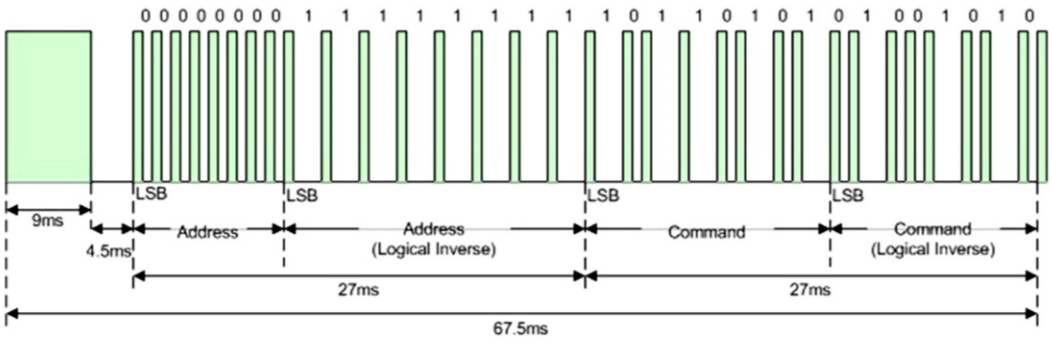


Figure 2.1: NEC Protocol at 67.5 ms [4]

Figure 2.1 shows the NEC protocol utilizes a pulse-distance encoding scheme where each frame consists of a leader code, address, and command bits. The total transmission time for a signal frame is approximately 67.5 ms.

2.2.2 Pulse Distance Width Modulation

Most AC manufactures utilize a variation of Pulse Distance Width Modulation. In this protocol, the total time of a bit (T_{bit}) is the sum of the pulse duration (T_{pulse}) and the space duration (T_{space}):

$$T_{bit} = T_{pulse} + T_{space} \tag{2.4}$$

2.2.3 Data Frame Structure and Checksums

AC IR frames are high-density, often exceeding 100 bits per transmission to communicate the full state of the device (Mode, Temp, Fan, Swing). To ensure the receiver ignores corrupted data, a Checksum (C) is often appended to the frame, calculated as the sum of the preceding data bytes (B) module 256:

$$C = \left(\sum_{i=1}^{n} B_i \right) \bmod 256 \tag{2.5}$$

This mathematical validation is critical for the ‘stateless’ automation proposed

in this study, as it ensures the AC unit only reacts to valid, uncorrupted commands [14].

2.2.4 Hardware Interfacing: The Optoelectronic Link

The physical interface consists of an IR LED and photodetector. The receiver module performs three critical functions: amplification, band-pass filtering at 38 kHz, and demodulation [15]. The relationship between the input light intensity (I) and the output voltage ($Vout$) of the internal pre-amplifier can be modeled as:

$$V_{out} = R_f \times \eta \times P_{in} \tag{2.6}$$

where:

R_f = feedback resistance, ohms (Ω);

η = responsivity of the photodiode, amperes per watt (A/W); and

P_{in} = incident optical power, watts (W).

2.2.5 Operational Mode Configurations in Air Conditioning Systems

The operational efficiency of an AC system is governed by the relationship between the mass flow rate of the refrigerant and the heat exchange capacity of the coils. The heat transfer rate (Q) during Cool Mode can be expressed by the general energy balance equation:

$$Q = m \times C_p \times (T_{in} - T_{out}) \tag{2.7}$$

where:

Q = heat transfer rate, watts (W) or BTU/hr;

m = mass flow rate of air, kilograms per second (kg/s);

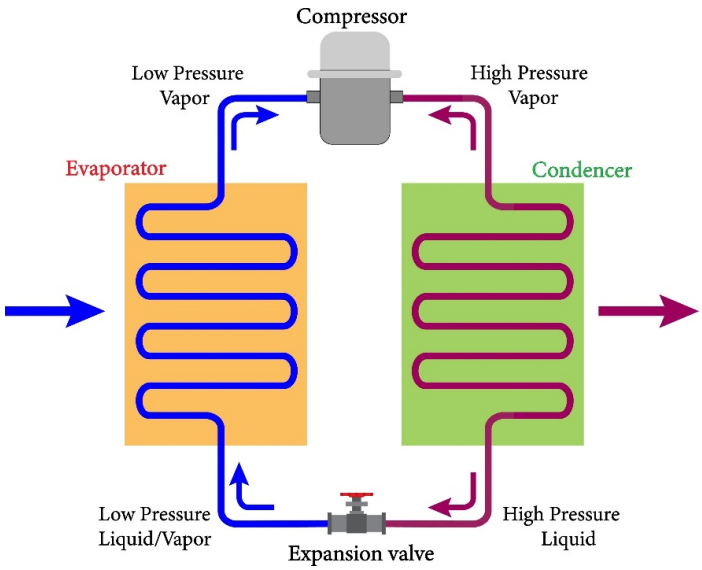


Figure 2.2: Vapor Compression Refrigeration Cycle [5]

C_p = specific heat capacity of air, joules per kilogram-kelvin (J/kg·K);

T_{in} = inlet air temperature, Kelvin (K); and

T_{out} = outlet air temperature, Kelvin (K). [5].

Figure 2.2 shows the Vapor Compression Refrigeration Cycle – represents the core thermodynamics process that enables Cool Mode in an air conditioning system. The cycle is a closed-loop system where a refrigerant undergoes phase changes to move heat from the indoors (heat source) to the outdoors (heat sink). The process consists of four primary stages governed by the laws of thermodynamics.

2.2.6 Dehumidification and Airflow Dynamics

In Dry mode, the system prioritizes latent heat removal over sensible heat removal. The efficiency of moisture extraction is linked to the bypass factor of the cooling coil. Conversely, Fan Mode operates purely on kinetic energy transfer, where power consumption (P) of the blower is proportional to the cube of the fan speed (N), according to the Fan Affinity Laws:

$$\frac{P_1}{P_2} = \left(\frac{N_1}{N_2}\right)^3 \tag{2.8}$$

where:

P = shaft power and

N = rotational speed.

This relationship highlights that Fan Mode serves as high-efficiency alternative for air circulation without thermal regulation [16].

2.2.7 Kinematic of Swing Mode

Swing Mode introduces a dynamic component to airflow distribution through the periodic oscillation of horizontal and vertical louvers. This mode is designed to eliminate “hot spots” and ensure a uniform temperature gradient across the zone. The angular velocity of the louver (Ω) and the resulting air throw distance are critical for achieving effective mixing. Mathematically, the distribution of air velocity (μ) at a distance (x) from the vent can be modeled using the turbulent jet theory:

$$\mu(x) = \frac{K \cdot \mu_0 \cdot \sqrt{A_0}}{x} \tag{2.9}$$

where:

K = center-line velocity constant;

μ_0 = initial discharge velocity; and

A_0 = effective area of the outlet. [17].

By varying the angle of the louver, the system alters the Coanda effect, either hugging the ceiling for long-range cooling or directing air downward for immediate localized relief.

2.2.8 Sensors

Sensor is a device that detects and responds to some type of input from the physical environment. In control monitoring, sensors are used to monitor temperature and humidity across the system, also Motion sensors are used to track various movements or motion for security purposes. The following are different types of sensors:

1. **Temperature sensors**

   The accuracy of an air conditioning system's setpoint maintenance is inherently limited by the sensitivity and response time of its temperature sensors. In engineering applications, these sensors are classified based on their material composition and electrical response to thermal excitation [21].

   1. **Thermocouples**: A passive sensor consisting of two dissimilar metal wires joined at one end (the sensing junction). They operate based on the Seebeck Effect, where a temperature gradient across the wires generates a small voltage (V).

      $$V = \alpha \left(T_{\text{hot}} - T_{\text{cold}}\right) \tag{2.10}$$

      where:

      α = seebeck coefficient;

      T_{hot} = hot junction temperature in K; and

      T_{cold} = cold junction temperature in K [6].

   2. **Resistance Temperature Detectors (RTDs)**: RTDs operate on the principle that the electrical resistance of pure metals increases linearly with temperature. They offer the highest accuracy and stability but are often

reserved for industrial-grade HVAC systems due to higher manufacturing costs.

3. **Thermistors**: The standard for residential AC units. As ceramic-based semiconductors, they exhibit a non-linear change in resistance. Negative Temperature Coefficient (NTC) thermistors are the industry standard, where resistance decreases as temperature rises [18].

2. **Humidity Sensors**

Humidity sensors, often referred to as hygrometers, are instruments intended to gauge the level of water vapor in the ambient atmosphere. Humidity is a crucial environmental factor in areas like HVAC (Heating, Ventilation, and Air Conditioning), industrial operations, and environmental assessment. The assessed metric is Relative Humidity (RH), which is the comparison of the existing water vapor in the air to the highest quantity of vapor the air can accommodate at specific temperature, which is shown as [22]:

$$RH = \frac{Ph}{P(T)} \times 100\% \tag{2.11}$$

where:

Ph = partial pressure of water vapor, and

$P(T)$ = saturation vapor pressure at temperature T.

1. **Capacitive Humidity Sensors**: the most prevalent sensor type in residential and commercial air conditioning units. These sensors consist of a substrate on which a thin film of polymer or metal oxide is deposited between two conductive electrodes. The sensing principle relies on the change in the dielectric constant (ε) of the polymer material as it absorbs water molecules [23]. The capacitance (C) is calculated using the

parallel-plate capacitor formula:

$$C = \frac{\varepsilon_r \varepsilon_0 A}{d} \tag{2.12}$$

where:

ε_r = dielectric constant;

ε_0 = permittivity of free space;

A = surface area of electrodes; and

d = distance between electrodes.

As ambient humidity increases, the polymer absorbs water vapor (H_2O, which has a high dielectric constant of approximately 80), thereby increasing the total capacitance. This change is then converted into a frequency or voltage signal by the system's signal conditioning circuit [24].

2. **Resistive Humidity Sensors (Hygristors)**: measure the change in electrical impedance of a hygroscopic medium, such as a conductive polymer or salt. Unlike capacitive sensors, resistive sensors exhibit an exponential relationship between resistance and humidity:

$$R = R_0 \cdot e^{-k\,(RH)} \tag{2.13}$$

where:

R_0 = base resistance;

k = material constant,

3. **Motion Sensors**

Motion sensors are devices designed to detect physical movement within a defined area. They are widely used in security systems, automation, energy

management, robotics, and human-machine interaction. In principle, motion detection involves identifying changes in physical properties such as infrared radiation, sound waves, electromagnetic fields, or image patterns.

1. **Passive Infrared (PIR) Motion Sensors**: PIR sensors detect motion by measuring changes in infrared radiation emitted by objects within their field of view. Humans and animals emit thermal energy in the 8-14 µm wavelength range. PIR sensor contains a pyroelectric material that generates an electric charge when it experiences differential heat patterns. Motion is detected when a warm body moves between segments of the sensor's Fresnel lens, causing fluctuation of infrared energy that the sensors convert into a voltage signal [26].
2. **Ultrasonic Motion Sensors**: Ultrasonic motion sensors emit high-frequency sound waves (typically around 40 kHz) and measure the time it takes for the waves to reflect off objects in the environment. Motion is detected by analyzing the Doppler shift or changes in the echo time-of-flight. When an object moves, the reflected ultrasonic waves experience a frequency shift that the sensor interprets as motion [27].
3. **Microwave Motion Sensors**: Microwave motion sensors operate by transmitting electromagnetic waves (typically at 2.45 GHz or 10.525 GHz) and measuring changes in the reflected signal. These changes occur due to the Doppler effect—moving objects alter the frequency of the returned signal. The sensor interprets these variations as motion [28].

2.2.9 Human Thermal Comfort

Thermal comfort refers to the state of mind in which individuals express satisfaction with the surrounding thermal environment, and it is one of the most essential parameters in HVAC (Heating, Ventilation, and Air Conditioning) system design.

According to the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE), thermal comfort is determined by a complex interaction of environmental and personal variables that influence the body's ability to maintain thermal balance [29]. The concept is fundamentally rooted in human thermoregulation, the physiological process by which the body maintains a core temperature around 37℃ despite environmental variations [30].

Thermal comfort is determined by 6 primary factors, categorized into environmental and personal variables:

1. Air Temperature

   Air temperature refers to the dry-bulb temperature of the surrounding air. It is one of the most influential comfort parameters because it directly affects heat transfer between the human body and the environment [31].

2. Relative Humidity

   Humidity influences the evaporative cooling of the skin. High humidity reduces sweat evaporation, decreasing thermal comfort, while very low humidity may cause dryness and skin irritation [32]. Optimal indoor humidity typically ranges from 40% to 60%.

3. Air Velocity

   Air movement enhances convective heat loss. Higher airspeed improves perceived cooling but may cause discomfort if excessive, such as draft feelings in cooler environments [29].

4. Mean Radiant Temperature (MRT)

   MRT is the uniform temperature of an imaginary surrounding enclosure where radiant heat exchange equals that of the actual environment. Because hu-

mans lose a significant portion of heat through radiation, MRT can substantially influence comfort even when air temperature is acceptable [33].

5. Metabolic Rate (Met)

   Metabolic rate describes the body's heat production, measured in met units (1 met $\approx 58\ \mathrm{W/m^2}$). Higher activity levels increase metabolic heat, requiring lower environmental temperatures for comfort [30].

6. Clothing Insulation (Clo)

   Clothing affects heat resistance between the body and environment. Insulation is measured in clo units (1 clo $\approx$ insulation of a typical business suit). A higher clo value reduces heat loss and increases warmth perception [29].

2.2.10 Regional Climate Constraints and Thermal Sensation in the Philippines

The Philippine archipelago is characterized by an average annual temperature of approximately 27℃ and relative humidity levels that frequently fluctuate between 70% and 85% [34]. These conditions often fall outside the "Static Comfort Zone" defined by Western standards, which typically recommend an indoor air temperature of 23℃ to 26℃ at 50% humidity. The high latent heat load in the Philippines significantly inhibits evaporative cooling—the body's primary mechanism for heat dissipation through sweating. The study of [35] suggests that because Filipinos are acclimatized to a hot-humid environment, their "neutral" temperature—the temperature at which they feel neither hot nor cold—is often higher than that of individuals in temperate climates, yet their preference in air-conditioned office environments tend toward the "cooler" side (21–24℃) as a psychological reprieve from the outdoor heat.

2.2.11 ANSI/ASHRAE Standard 55 Framework

The primary regulatory and technical benchmark for indoor environments is the ANSI/ASHRAE Standard 55, titled 'Thermal Environmental Conditions for Human Occupancy.' This standard specifies the combinations of indoor thermal environmental factors and personal factors that will produce conditions acceptable to a majority ($\geq 80\%$) of the occupants [36]. As updated in the 2023 revision, the standard systematically addresses six core variables: four environmental factors (air temperature, radiant temperature, air speed, and humidity) and the two personal parameters (metabolic rate and clothing insulation). The standard's methodology is bifurcated into two primary approaches: the analytical (PMV/PPD) Method, intended for mechanically conditioned buildings, and the adaptive model, which is applicable to naturally conditioned spaces where occupants have a greater degree of control over their environment [37].

2.2.12 ANSI/ASHRAE Standard 55 Framework

PMV is a mathematical index that predicts the mean value of the thermal sensation votes of a large group of people on a psycho-physical scale [38]. The index is derived from the balance between the internal heat production of the body and its heat loss to the environment. PMV is expressed as a value between -3 and +3, where the integers correspond to specific sensations.

+3: Hot

+2: Warm

+1: Slightly Warm

0: Neutral (Comfort)

-1: Slightly Cool

-2: Cool

-3: Cold

2.2.13 Predicted Percentage Dissatisfied (PPD) Index

The Predicted Percentage of Dissatisfied (PPD) index estimates how many people are likely to feel thermally uncomfortable in each environment. Rather than describing sensation, PPD expresses discomfort as a percentage. Even in an ideal environment (PMV = 0), the PPD will never reach 0%; the minimum is 5%, meaning at least some occupants will always feel slightly warm or slightly cool. PPD increases as the PMV deviates from zero; for example, PMV values near ±1 may correspond to 20% dissatisfaction. Because of this, PPD is useful for setting allowable comfort ranges, such as ASHRAE Standard 55, which typically requires a PPD below 10% for acceptable comfort. PPD provides building designers with a quantitative measure of discomfort risk, helping guide HVAC control strategies to enhance occupant well-being [39, 40].

2.3 Literature Review

This section examines the different applications in innovating the control systems of an air conditioning unit by integrating hardware innovations, software applications as well as IoT integration. The findings highlight the potential of using automatic control system analogy for energy efficiency and thermal comfort for the users.

2.3.1 Manual Remote-Control System

Recent studies [41, 42, 43, 44] emphasize scenarios where users still perform control actions such as changing setpoints and modes, while systems focus on im-

proving decision quality, characterizing behavior, or enabling more efficient manual interfaces.

The study in [41] characterizes AC IR remote control as a brand-specific, line-of-sight communication channel where ON/OFF, temperature, and mode changes are encoded in fixed data frames, emphasizing compatibility and feedback limitations of conventional manual remotes by the used of longitudinal thermal imaging.

The study in [42] used CatBoost model and an intelligent monitoring platform to predict AC power consumption and indoor temperature, enabling cost-optimal setpoint scheduling that reduces energy use and operating cost while respecting manually defined comfort bounds.

Study in [43] used YOLOv5-based occupancy detectors to divide a classroom into zones and control AC outlets per occupied zone, reducing energy consumption by around 21-22% while raising issues of privacy and deployment complexity for camera-based control.

The study in [44] used by combining laser-photoresistor counting, PIR sensing, and NRF24L01 wireless links to estimate occupancy and heat load, switching AC units ON/OFF accordingly, achieving notable energy savings while still relying on manual temperature setpoint adjustments and facing limitations in sensor placement and classification accuracy.

2.3.2 Automatic Remote-Control Systems

Automatic AC RRL centers on IoT-based systems that use environmental and occupancy sensing to directly actuate AC modes, setpoints, and schedules without continuous user input. Typical inputs include PIR motion sensors, temperature and humidity sensors (DHT11/DHT22, LM35), and sometimes current sensors and cameras, with wireless links via Wi-Fi, ZigBee, or web interfaces. These works commonly modulate IR commands (around 38 kHz) toward conventional indoor

units, showing that legacy ACs can be converted into smart, automatically controlled devices at low cost.

The study in [45] used a PIR motion sensor and a DHT11 temperature sensor to automatically switch the AC ON/OFF and select discrete comfort levels (cold/medium/cool), demonstrating low-cost IoT-based reduction of AC control errors.

Study in [46] employs an IR module and DHT11 sensor with a Blynk mobile interface to provide remote ON/OFF and swing control, highlighting reliable operation within 1-4m and dependence on Wi-FI signal quality.

The study in [47] developed a Smart Remote for IoT-Based Air conditioner using ESP32 with a Telegram bot to support remote manual ON/OFF via chat commands, successfully reproducing stored brand-specific IR codes but remaining limited by hardcoded protocols and Wi-Fi connectivity.

The study in [48] developed a web-interface AC controller using temperature/humidity sensing and ZigBee–Wi-Fi connectivity, enabling remote ON/OFF and temperature control while being constrained by internet speed and ZigBee range.

Study in [49] trained a YOLOv5 model on thermal images for binary occupancy detection, achieving high precision and recall suitable for privacy-preserving HVAC control, although tested with a single participant and limited viewpoints.

The study in [50] developed a smart remote linked to Blynk that automatically increases or decreases AC temperature around 25℃ setpoint using room temperature feedback but does not incorporate occupancy detection or other AC modes.

The study in [51] designed an “Enhanced Air Conditioning Control for the Smart Home”. This work connects an ESP32 directly to the AC CN105 service port to provide web-based control of AC settings without additional sensors, offering low-cost integration but only for compatible units and requiring manual hardware modification.

Study in [52] used a system of DHT22 temperature and humidity sensor and PIR-based presence detection with WiFi networking to automatically switch AC and adjust valves, stabilizing temperature and humidity while depending on short-range IR and network stability.

The study in [53] used an IoT platform with temperature, motion, current, light, and camera inputs coordinates AC, fan, and lighting, automatically adjusting operation based on presence and temperature and logging attendance, though limited by WiFi coverage and simple temperature control logic.

Study [54] used DHT22, PIR, and IR transmission with Blynk interface, the system alternates AC and fan operation according to motion and temperature, achieving substantial energy and cost reductions while constrained by IR line-of-sight and WiFi range.

Overall, these studies show that sensor-and IoT-based systems can be independently switch ACs and fans, tune setpoints, and coordinate other devices based on occupancy and environmental conditions, which reduces energy use and user effort while maintaining acceptable comfort. Manual control systems keep users in charge of setpoints and modes but add intelligence through machine learning, occupancy sensing, and detailed IR/thermal analysis to guide more efficient usage and achieve substantial energy and cost savings without fully automating the decision process.

Chapter 3

RESEARCH METHODOLOGY

This chapter covers the development of both the hardware and software components of this study, detailing the methods of execution. Additionally, this chapter highlights the application of these proposed methods through experiments.

3.1 Proposed Method

This project is a smart, room-responsive remote-control system for air conditioning units. The proposed model mainly consists of sensors as input devices such as temperature sensors, humidity sensors, and motion sensors. The system uses an ESP32 microcontroller unit for logical decision making. The main system attached with a momentary switch for voting mechanism, and an IR transmitter in sending execution command to the aircon.

Figure 3.1 shows the simplified block diagram of the smart-remote control system for this study. As shown, the primary components of the system encompass sensors, control processing, communication, and power supply. Sensors including temperature, humidity and motion sensors monitor and read ambient room conditions and transmitted them to the control hub for logic processing. After the system processes all the data, the control will send a command to the air conditioner to change its mode and settings based on the gather data from the sensors.

3.1.1 System Components

The following are the actual components used in AIRWise system:

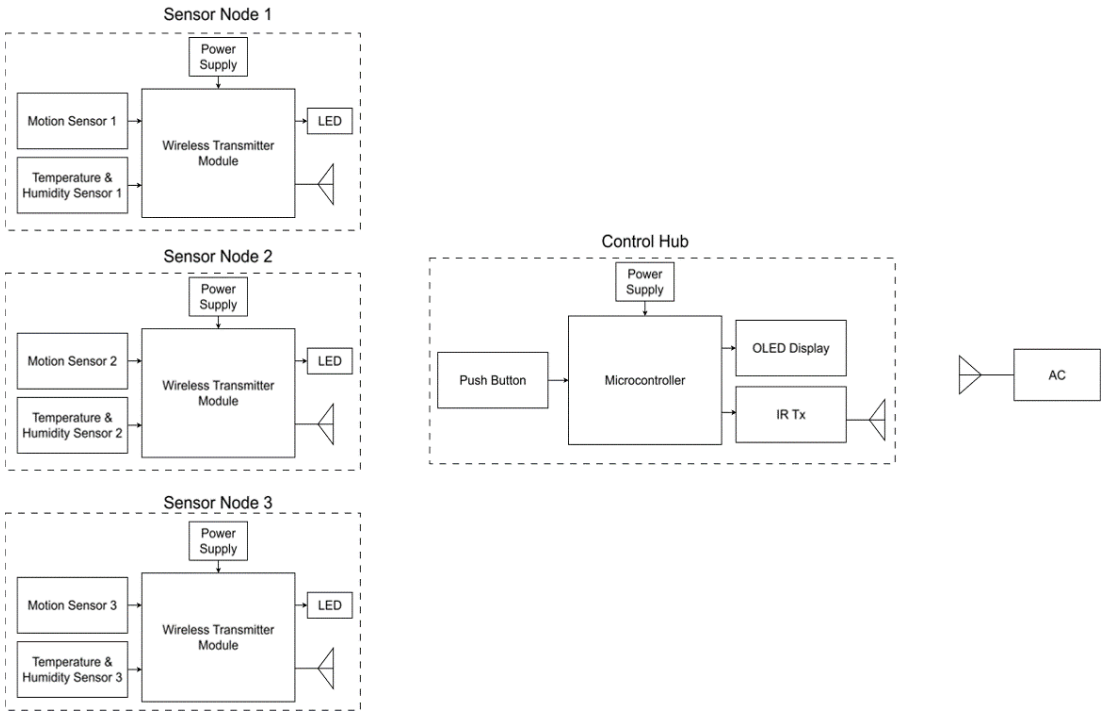


Figure 3.1: Simplified Block Diagram of the System

Input Devices

1. Temperature and Humidity Sensor

   Figure 3.2 is a DHT22 temperature and humidity sensor, which operates on 3.6 V to 5 V power supply range. It can measure from 20% to 90% humidity range and 0℃ to 50℃ temperature range, with a humidity accuracy of +/-5% and +/-2℃ accuracy for temperature. This is used to measure the temperature and humidity in the system.

2. PIR Motion Sensor

   Figure 3.3 is a HCSR501 PIR motion sensor, which operates on 5 V to 20 V power supply range. It has a sensing range of less than 120 degrees, within 7 meters, TTL output of 3.3 V and 0 V. The trigger methods, L-disable repeat trigger, and H-enable repeat trigger. This is used to measure motions in the system.

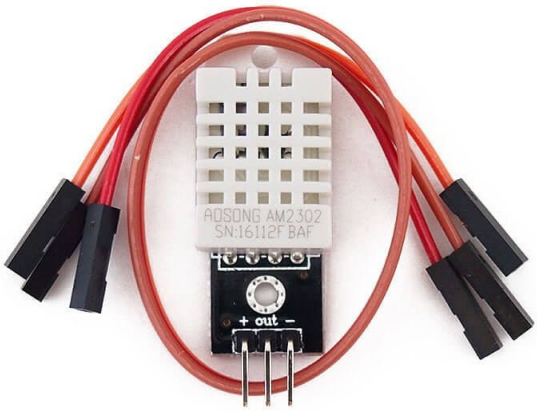


Figure 3.2: DHT22 temperature and humidity sensor



Figure 3.3: HCSR501 PIR motion sensor



Figure 3.4: Self-reset switch



Figure 3.5: 940 nm IR LED

3. Push Button Switch

   Figure 3.4 is a DSZUJTYROP Polyamide self-reset switch, which operates up to 50 V. The contact type is a momentary (normally open) that can be self-reset after it was pressed. It has 1,000,000 cycles of mechanical life and has a contact bounce of 10 ms. The switch is designed for user's voting input.

4. Infrared Transmitter

   Figure 3.5 is a 940 nm IR LED, which operates from 1.2 V to 1.4 V power supply range. It has a peak wavelength of 940 nanometers; it has a viewing angle of 30 degrees and a radius flux of 40 milliwatts. The purpose of IR LED is to send encoded IR signals to the air conditioner.



Figure 3.6: 1.3” OLED Display I2C

5. OLED Display I2C

   Figure 3.6 shows a 1.3-inch I2C OLED Display, which operates at 3.3 V to 5 V supply voltage range. The display has a resolution of 128 x 64 pixels and communicates using I2C serial protocol, requiring only SDA and SCL data lines. It utilizes organic light-emitting diode (OLED) technology, allowing each pixel to emit its own light, resulting in high contrast, low power consumption, and wide viewing angles. The purpose of the OLED display is to visually present system information such as temperature, mode settings of the air condition of the control system.

Microcontroller Unit

The microcontroller processes and stores the signals generated by each sensor and analyzes the gathered data via logical processing and sends it an execution command to the air conditioner. The microcontroller used in this study is the ESP32, as shown in Figure 3.7. The ESP32, developed by Espressif Systems, is a multifunctional microcontroller equipped with Xtensa Dual-Core processor, integrated Wi-Fi and Bluetooth, and a comprehensive set of peripherals, including GPIO, ADC, DAC, PWM, and standard communication interfaces like I2C, SPI, and UART. It offers various power management modes suitable for battery-operated devices and includes security features such as hardware encryption and

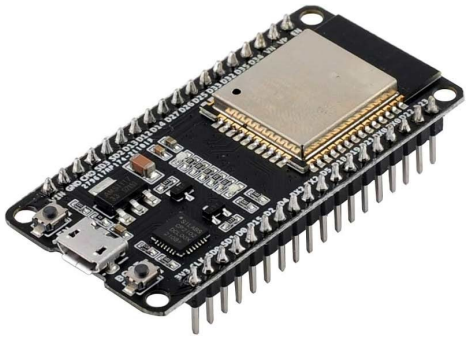


Figure 3.7: ESP32 Microcontroller

secure boot. With support from development environments like Arduino IDE and Espressif's ESP-IDF, the ESP32 is perfect for IoT applications, home automation, wearable technology, and industrial automation, delivering strong performance and versatility for numerous uses.

1. ESP8266 Node MCU

   Figure 3.8 is a Node MCU ESP8266 is a compact WiFi development board built around the ESP8266EX SoC, which includes a 32-bit processor, RAM, and a 2.4 GHz 802.11 b/g/n transceiver. It typically runs at 80 Mhz to 160 Mhz, with about 4 MB flash and a 10-bit ADC and exposes around 16 GPIO pins that support digital I/O, SPI, I2C, PWM, and UART for sensor and actuator interfacing. Operating at 3.3 V logic and powered via micro-USB or VIN through an on-board regulator, it integrates a full TCP/IP stack with station, access-point, and AP+station modes plus WPA/WPA2 security. It is used as an MCU to temperature, humidity, and motion sensors that store the data and transmit it to the ESP32.

Output Devices

1. Panasonic Window-type Inverter AC

   Figure 3.9 shows the Panasonic CW-XU1021VPH window-type air conditioner, which serves as the controlled appliance of the system. It operates

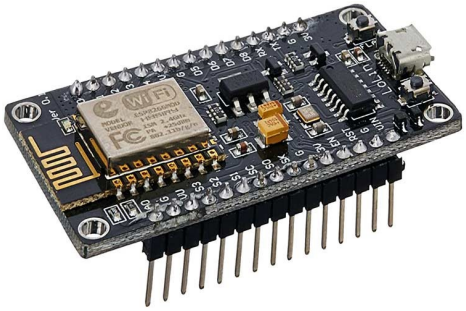


Figure 3.8: ESP8266 Node MCU



Figure 3.9: Panasonic CW-XU1021VPH window-type air conditioner

using a single-phase AC power and supports infrared (IR) remote control for functions such as power, temperature, fan speed, and operating modes. In this study, the unit receives encoded IR commands transmitted by the proposed smart remote-control system to achieve automated thermal comfort control.

3.1.2 Prototype

3D Design

Figure 3.10 shows the 3D design of the 3-sensor node enclosure with approximately 120 mm x 60 mm x 25.5 mm. The enclosure includes the internal mounts for the MCU and sensors, ventilation slots near the temperature and humidity sensors for air exposure, and a 12 mm front aperture for the motion sensor, Side opening

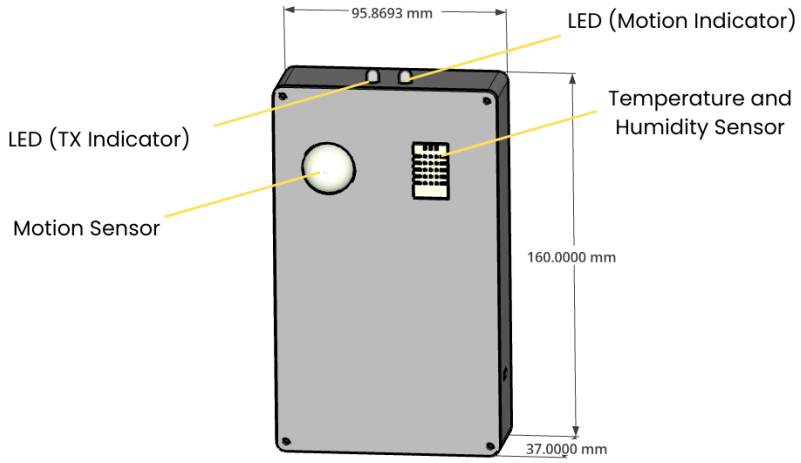


Figure 3.10: Sensor Node

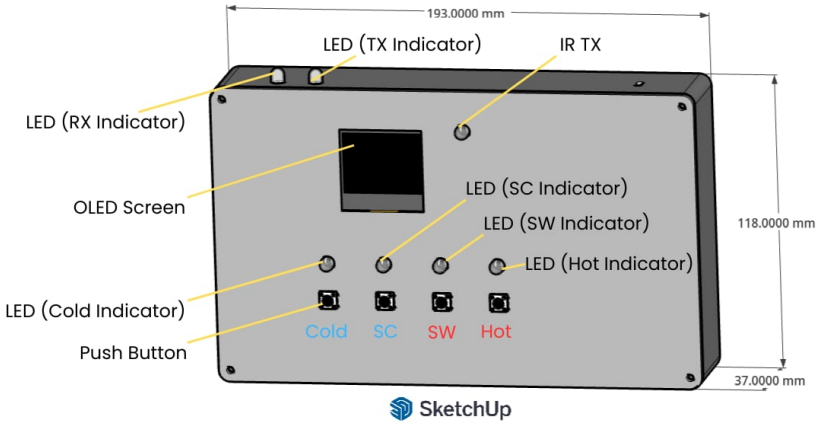


Figure 3.11: AIRWise Control Hub

are provided for power and communication ports.

Figure 3.11 shows the 3D design layout of the control hub, housing an ESP32 MCU mounted on internal standoffs. The front panel integrates a 1.3-inch I2C OLED Display centrally positioned for visibility, with four tactile push buttons arranged below it for user interaction. A circular aperture on the front edge is provided for the 940 nm IR tx to ensure a clear LOS toward the air-conditioning unit. Side openings are allocated for power and communication ports.

3.1.3 Algorithm

Figure 3.12 shows the flow of the system of the three sensor nodes. The system operates in a continuous loop designed for data collection, processing, and wireless reporting. It begins by initializing the control layer with a fixed header value of 0x01 and then enters its main cycle. First, the system captures raw inputs from motion, temperature and humidity sensors. The raw values for the temperature and humidity are immediately scaled by dividing them by 16 to ensure they are in the correct format for the system. Once the data is refined, it is bundled into a sensor data array. To prevent data corruption during transit, the system runs a Cyclic Redundancy Check (CRC) to generate an error-detection code. This code is appended to the control ID and sensor data to form a final packet (Sn1). This complete package is then transmitted via WiFi to the Control Hub, after which the system instantly loops back to begin the next reading.

Figure 3.13 shows the Cyclic Redundancy Check (CRC) subroutine to ensure the sensor data isn't corrupted during transmission. The system begins by initializing a loop to process the three specific bytes $(N = 3)$, starting with an XOR operation between the current CRC value and the sensor input. Inside this process, an inner loop runs eight times to evaluate the byte bit-by-bit; if the Most Significant Bit is active (1), the system performs a left shift followed by an XOR with the polynomial 0x07, otherwise, it simply shifts the value. Once all bits and bytes are cycled through, the final CRC_{rec} is returned to be appended to the transmission packet for the Control Hub.

Figure 3.14 shows the flow of the control hub which acts as the central intelligence of the system. It begins by initializing a counter (M=0) and a motion threshold voltage (Mth = 2.5V) before entering a loop to receive data packets from three separate sensor nodes. If the detected motion meets or exceeds the 2.5 V threshold, the system logs the event to a database and increments the counter.

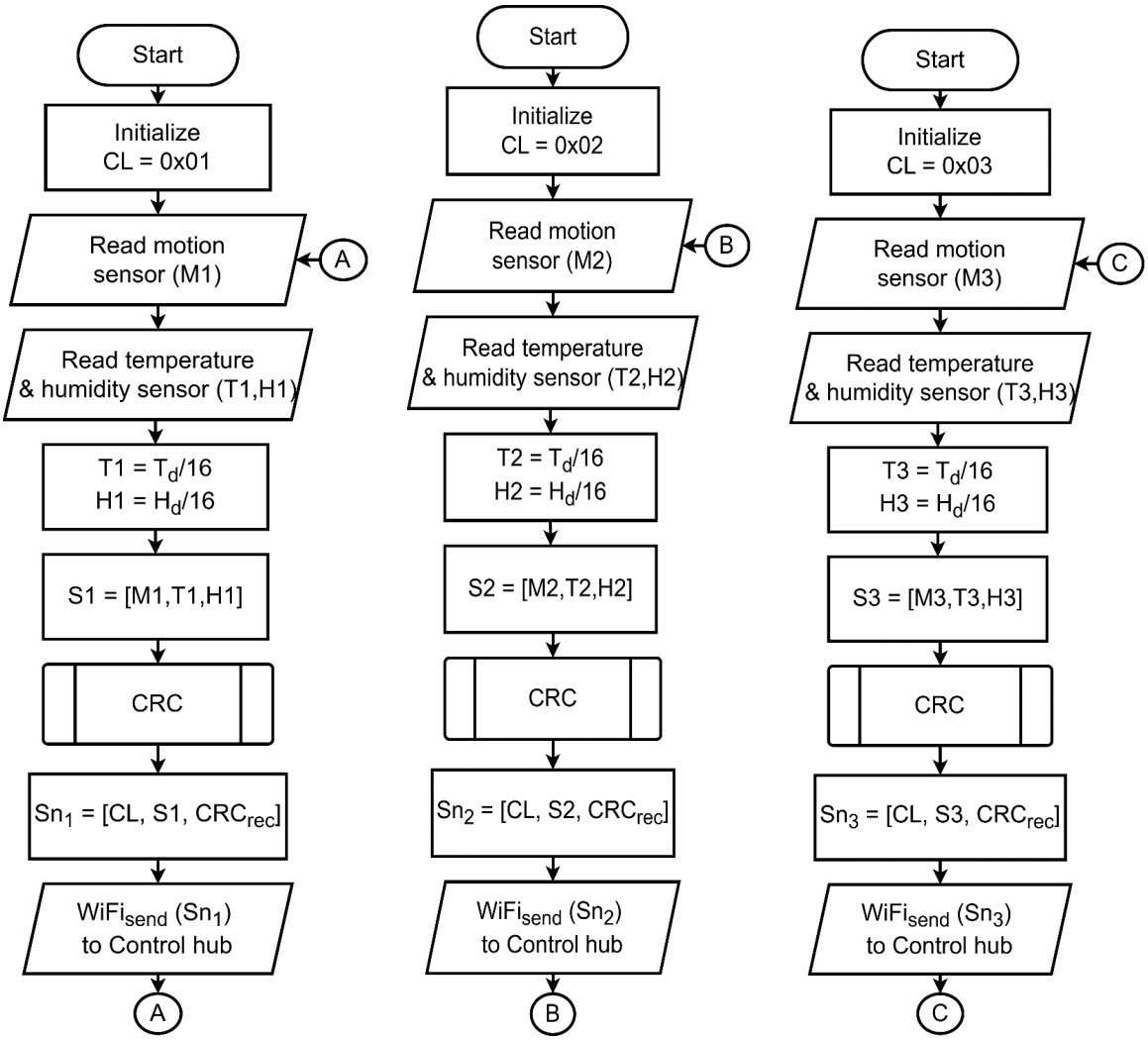


Figure 3.12: Sensor Nodes 1,2,3 Flow

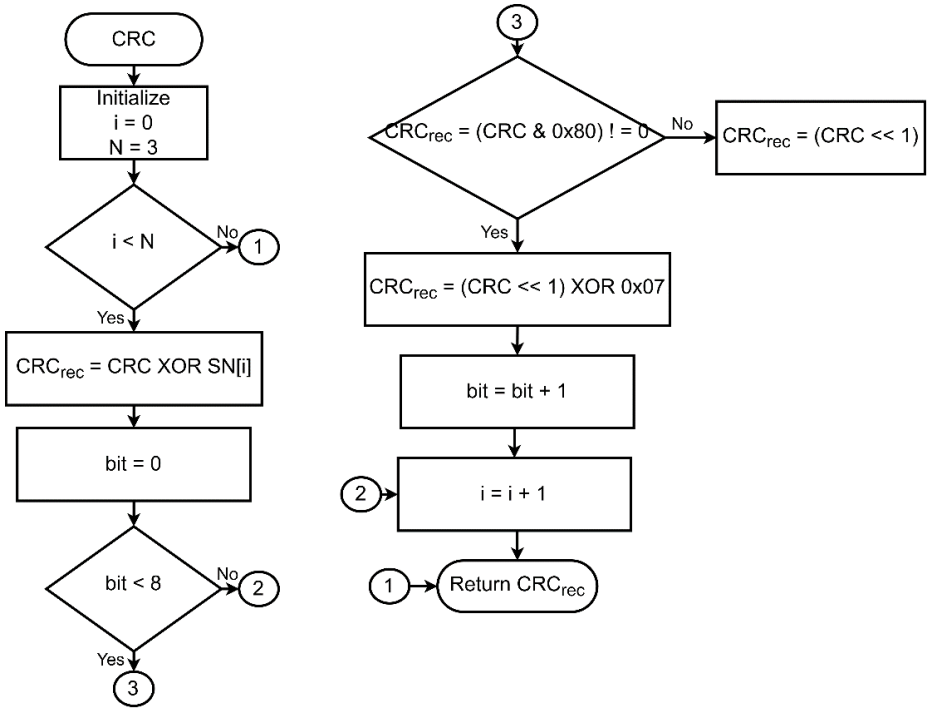


Figure 3.13: CRC_{rec} Subprocess

Once 30 valid motion events are recorded, the hub transitions to an advanced decision-making phase at point D. In the second phase, the system executes several subroutines to determine how to adjust the environment. It evaluates mode and temperature selections, manages swing timing, and employs a voting mechanism to reconcile data from three sensor nodes. It also performs a CRC check to verify data integrity and runs error handling protocols to ensure system stability. Finally, the hub generates a specific text command string containing the target temperature, mode, and power status-and transmits it as an IR command to the aircon before concluding the cycle.

Figure 3.15 shows the predefined process for mode selection and relative temperature. After receiving the temperature readings from the three sensor nodes and the aggregated humidity value, the control hub first establishes a reference temperature, which corresponds to the current air-conditioning set temperature if the unit is already operating, or a predefined comfort reference when the air-

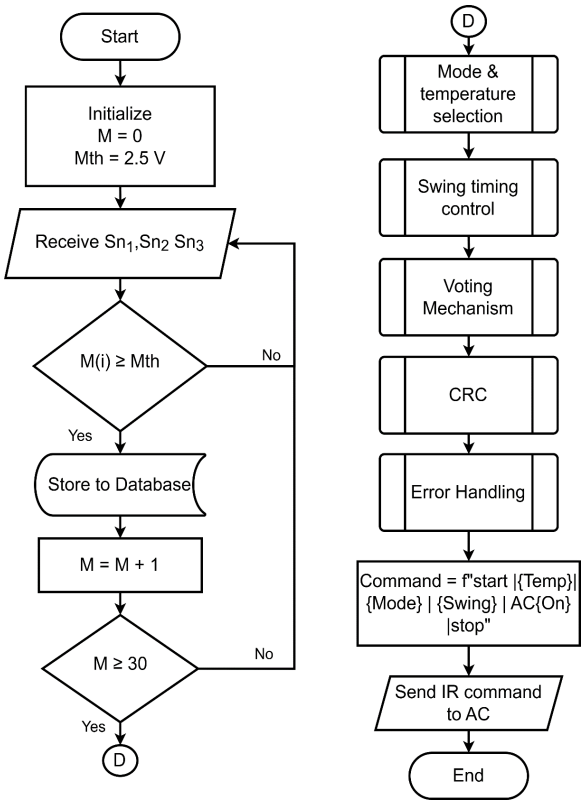


Figure 3.14: Control Hub Flow

conditioning is initially off. Each sensor node's temperature is then compared against this reference using a predefined tolerance range to classify the local thermal condition as hot, cold, or within the comfort zone. The control hub evaluates the combined thermal states of the three nodes using a majority rule to determine the dominant room condition.

Based on this evaluated thermal condition and the measured humidity level, the control hub selects the appropriate operating mode. If the humidity exceeds the defined threshold, the system prioritizes dry mode to reduce moisture levels. If humidity is within acceptable limits and the dominant thermal condition is hot, cooling mode is selected. Conversely, if the room is classified as cold or within the comfort range, the system selects fan mode to maintain air circulation without active cooling. Once the operating mode is determined, the relative temperature adjustment subroutine calculates the required change to the air-conditioning

temperature setting. If the dominant thermal state is hot, the set temperature is decreased incrementally, whereas if the room is cold, the set temperature is increased incrementally. If the room is within comfort range, the current temperature setting is maintained. The final temperature value is constrained within the allowable operating range of the air-conditioning unit before the control hub generates and transmits the corresponding control commands.

Figure **??** shows the swing control predefined process. After receiving temperature data from the three sensor nodes, the control hub identifies the hottest zone (left, middle, or right) through relative comparison of the sensor readings. The system then checks whether the air-conditioning unit is turned on before executing any swing action. Because the air-conditioning unit does not provide feedback on vane positioning, swing positioning is implemented using a time-based method. A full left-to-right swing is calibrated to take eight seconds, allowing specific time offsets to represent the left, center, and right positions. The control hub calculates the required movement time based on the difference between the target position and the last stored vane position. It then activates the swing motor via an infrared command and stops it after the calculated duration, causing the vane to face the hottest zone. The final vane position is stored in memory to ensure consistent airflow direction during subsequent operation cycles.

Figure **??** shows the error-handling routine, by initializing all system parameters and limits, including the temperature readings from each sensor node T_i, allowable temperature bounds $(TC_{\min}, TC_{\max})$, maximum retry count $N_{\max}$, maximum allowable temperature change $\Delta T_{\max}$, and voting threshold ΔT_{vote}. The system then iterates through each sensor node starting with $i = 1$, acquiring the current temperature reading T_i. If the transmission-valid flag $T_{(v,i)}$ indicates invalid data, the retry counter $N_{(r,i)}$ is incremented, and the system attempts to reacquire the data until either valid data is received or the retry limit is exceeded, in which case the

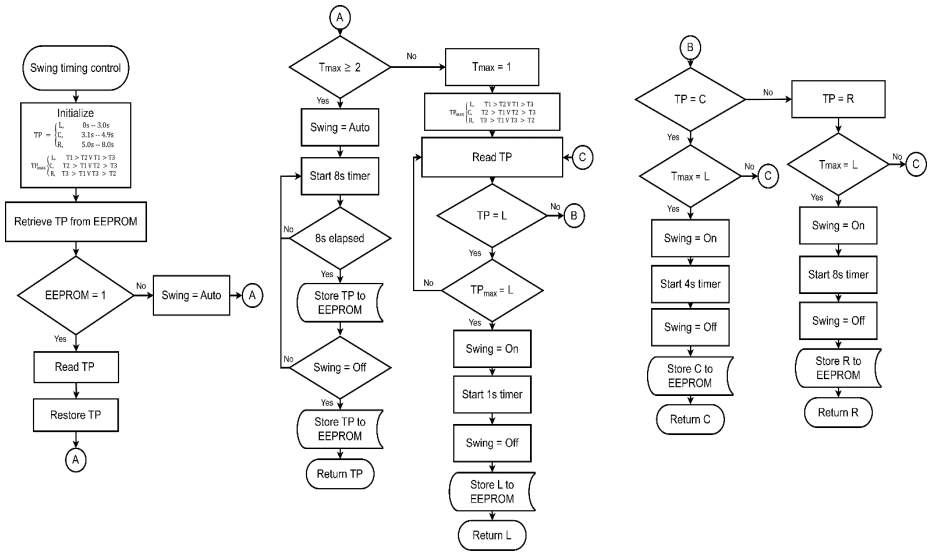


Figure 3.15: Swing control subroutine

sensor is flagged as INVALID. Once valid transmission is confirmed, the temperature reading is checked against the allowable physical range; readings outside $TC_{\min}$ and $TC_{\max}$ are immediately marked INVALID. For readings within range, the system performs temporal validation by comparing the current temperature $T_i(t)$ with the previous value $T_i(t-1)$; if the change exceeds $\Delta T_{\max}$, the sensor is again flagged INVALID. Additional integrity checks follow, including CRC validation and a communication-link check, both of which must pass for the reading to be accepted. Valid readings are stored, and the system proceeds to the next sensor until all nodes have been processed. Afterward, the number of valid sensors N_{valid} is counted; if fewer than two valid sensors remain, the system declares an error and displays it on the OLED. If at least two sensors are valid, a voting process compares each valid temperature against the computed vote temperature T_{vote}; any sensor whose deviation exceeds ΔT_{vote} is considered an outlier and excluded. The system then reassesses N_{valid}; if at least two valid sensors still remain and the system status flag S_{sys} is normal, control returns to the main program. Otherwise, the system signals an error on the OLED display, ensuring that unreliable or inconsistent temperature data never propagates into the control logic.

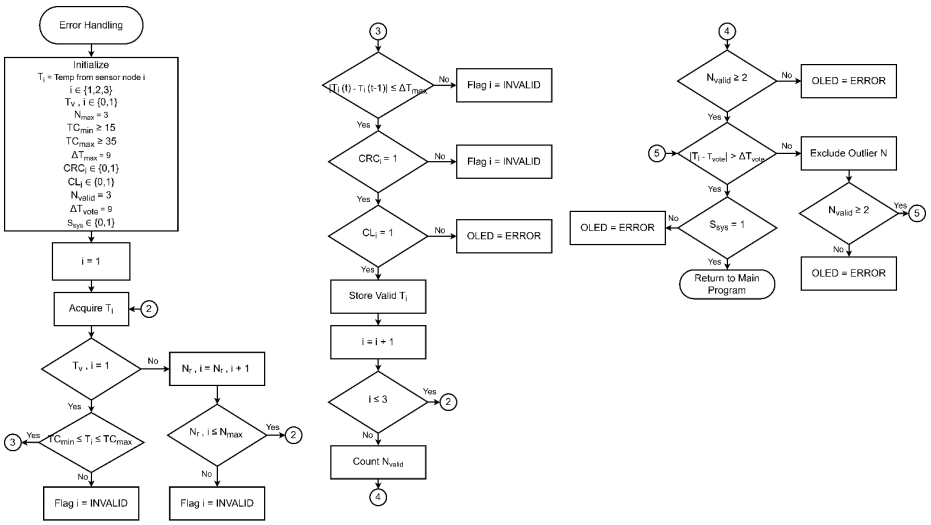


Figure 3.16: Error-handling subroutine

Figure 3.17 shows the idle voting mechanism process. Each temperature reading is independently classified as hot, slightly warm, slightly cool, and cold by comparing it to the reference temperature within a defined tolerance range. The control hub then counts how many sensor nodes fall into each category and apply a majority rule. If at least two nodes indicate the same thermal state, that state is selected as the overall room condition. This approach prevents control decisions from being influenced by a single outlier sensor and ensures more stable and reliable air-conditioning operation.

3.1.4 Circuit Configuration

Figure 3.18 shows the actual full circuit connection of all the components of the sensor nodes. The DHT22 temperature and humidity sensor is connected to the digital GPIO pin D4 of ESP8266 node MCU. The power pin of the DHT22 temperature sensor is connected to the 3V pin of the ESP8266. The GND of the DHT22 is connected to the GND of the ESP8266. The HCSR501 PIR motion sensor is connected to the GPIO AO of the ESP8266 node MCU. The power pin of the HCSR501 sensor is connected to the 3V pin of the ESP8266. The red LED is

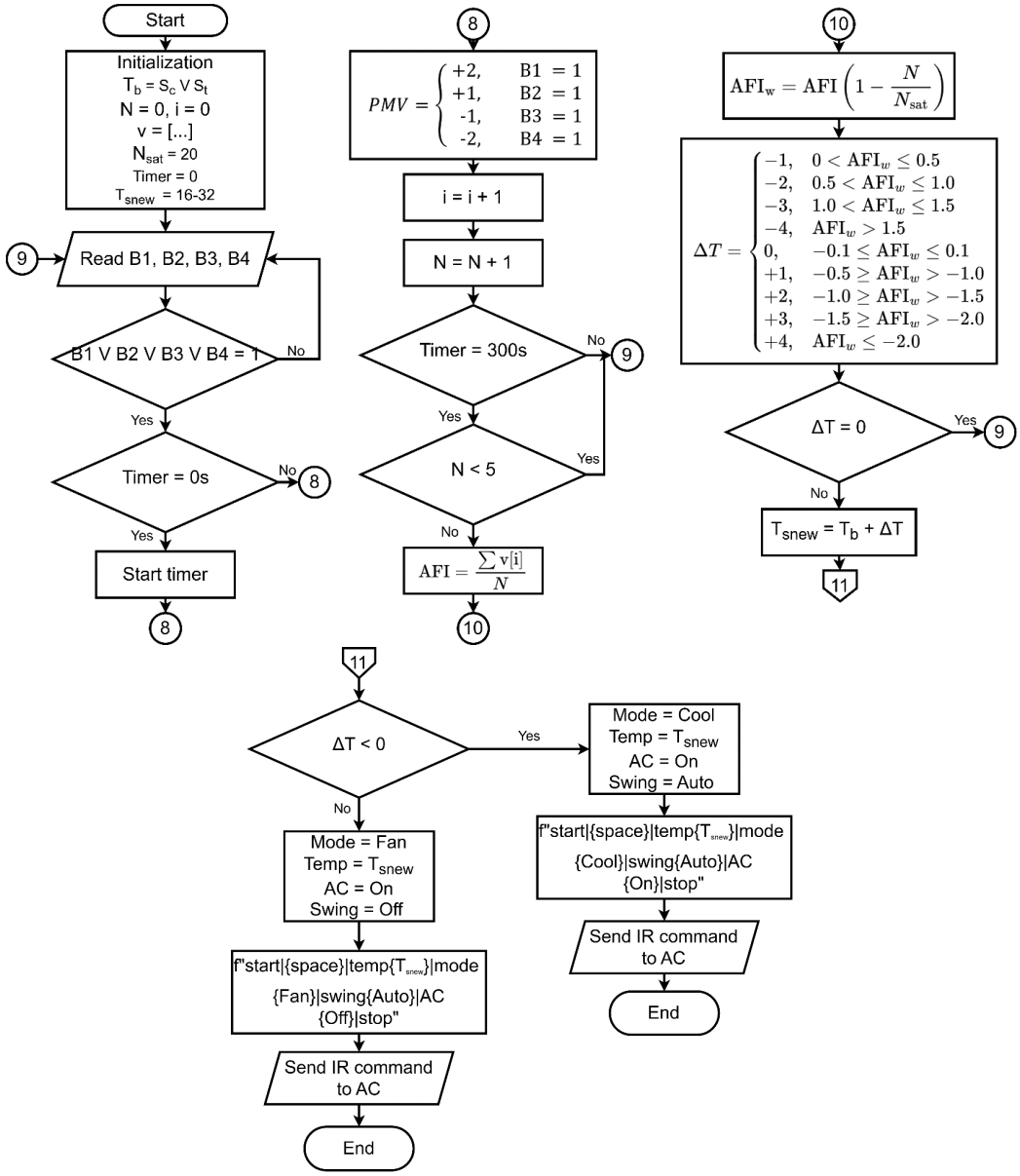


Figure 3.17: Voting mechanism flow

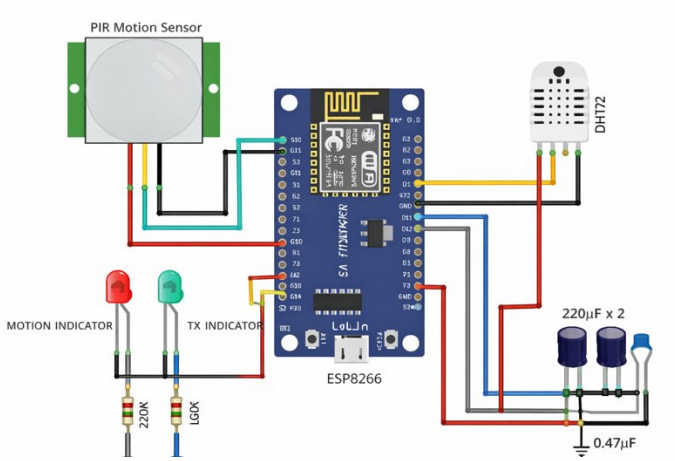


Figure 3.18: Sensor Nodes 1,2,3 Schematic Diagram

connected to the GPIO D6 of the ESP8266 node MCU. A 220 Ohms pull-up resistor is placed between the data and the GPIO D6 to avoid configuration floating. The green LED is connected to the GPIO D5 of the ESP8266 node MCU with a 100 ohms pull-up resistor. The two 220 microfarad electrolytic capacitors and the 0.47 microfarad ceramic capacitors are both connected to Vin of the ESP8266 node MCU to stabilize the voltage, filter noise, and provide temporary power bursts of the system.

Figure 3.19 shows the full schematic diagram for the control hub. The OLED I2C power is connected to the Vin of the ESP32. The SCK of the OLED I2C is connected to the SCK of the ESP32. The SDA of OLED I2C is connected to the SDA of the ESP32. The 940 nm IR transmitter is connected collector side of the 2N222 transistor. The base of the 2N222 is connected to the GPIO 4 of the ESP32 with a 47 ohms pull-up resistor. The emitter of the 2N222 is connected to the GND of the ESP32 with 22 ohms. The 4 tactile pushbuttons are connected to the GPIO 27, GPIO 14, GPIO 13, and GPIO 12 of the ESP32. The ground of the 4 tactile push buttons is all connected to the GND of the ESP32 with 1k ohms of pull-up resistor each. The blue LED is connected to GPIO 26 of the ESP32 with 47 ohms of pull-up resistor. The green LED is connected to GPIO 25 of the ESP32 with 100

ohms of pull-up resistor. The yellow LED is connected to GPIO 33 of the ESP32 with 100 ohms of pull-up resistor, and the red LED is connected to GPIO 32 of the ESP32 with 220 ohms of pull-up resistor. The green LED for the IR indicator is connected to GPIO 17 with 100 ohms of pull up resistors. The red LED of the IR indicator is connected to GPIO 16 with 220 ohms of pull-up resistor. The two 220 microfarad electrolytic capacitors and the 0.47 microfarad ceramic capacitors are both connected to Vin of the ESP8266 node MCU to stabilize the voltage, filter noise, and provide temporary power bursts of the system.

Figure 3.20 shows the full schematic diagram for the IR receiver for AC representation. The OLED I2C powers is connected to Vin of the ESP8266 12-E. The SCK of the OLED I2C is connected to the GPIO16 of the MCU, the SDA of the I2C is connected to the GPIO14 of the microcontroller. The TSOP 4838 IR receiver out pin is connected to GPIO1 of the microcontroller, the VCC are connected to power supply of the microcontroller, and the GND of the IR receiver are connected to the GND of the microcontroller.

3.2 Description of Experiments

This section contains detailed information on the experiments essential for the study.

3.2.1 Temperature and Humidity Sensor Accuracy Test

The objective of the experiment is to measure the accuracy of the DHT22 digital temperature and humidity sensor.
The materials are:

1. DHT22 digital temperature and humidity sensor;
2. Thermometer;

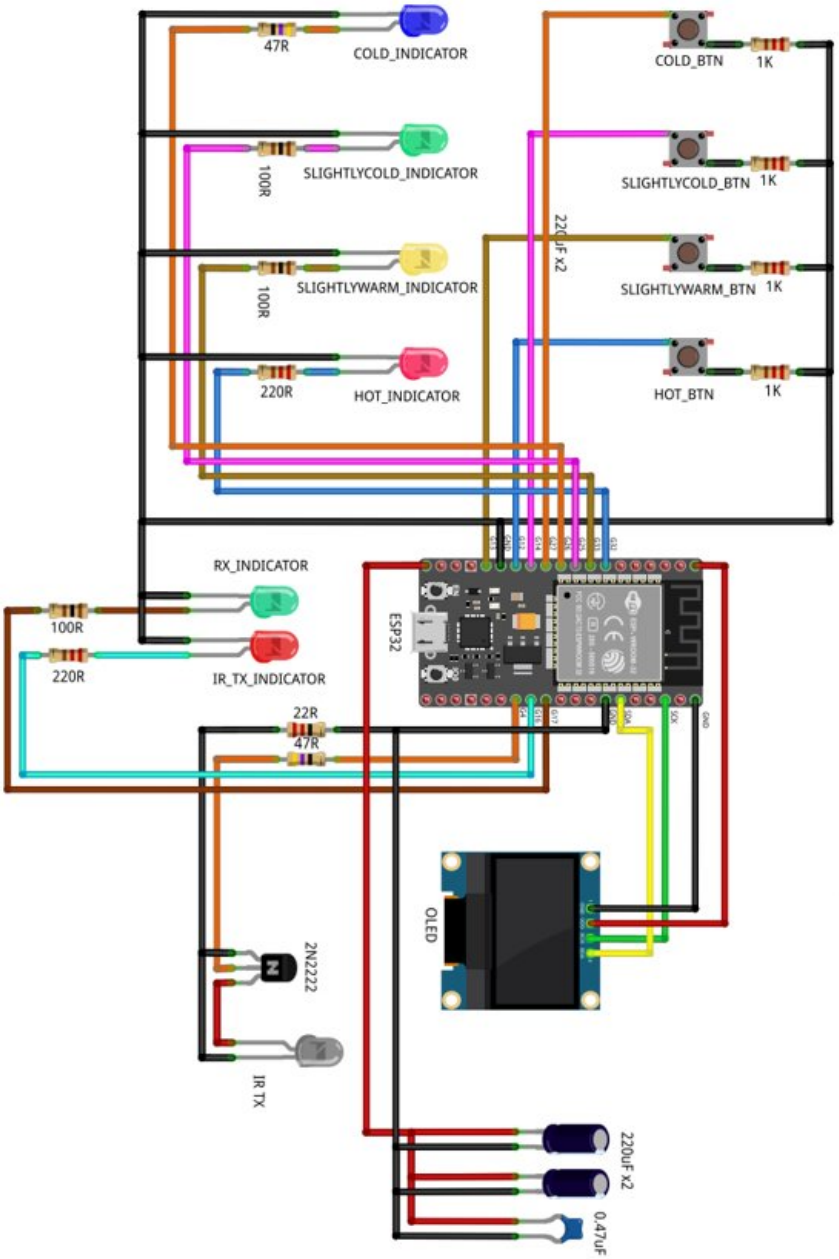


Figure 3.19: Control Hub Full Schematic Diagram

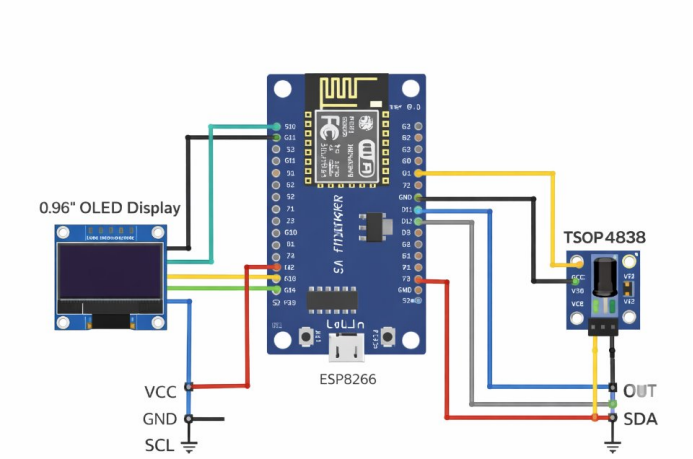


Figure 3.20: IR receiver schematic Diagram

3. Hygrometer;
4. 5m x 7m air-conditioned classroom;
5. DHT22_Calibration.ino;
6. Panasonic Window Type air-conditioning unit; and
7. Ruler.

The procedure is as follows:

1. Prepare the workstation and the materials needed.
2. Prepare the software for the DHT22 accuracy test:
    a. Power on the laptop.
    b. Once the laptop is powered on and logged in, locate the Arduino IDE software icon on the desktop and double-click it to open.
    c. In the Arduino IDE, go to File Tab, click it and navigate to the project folder, and select the file named "**DHT22_calibration.ino**".
    d. Connect the ESP8266 Node MCU with DHT22 sensor to the laptop using a USB data cable.

e. In the Arduino IDE, go Tools then Port and verify that the laptop recognizes the connected COM port.

3. Upload the calibration firmware:

   a. After the ESP8266 Node MCU successfully connected to the laptop.

   b. Click the right arrow icon (Upload) in the Arduino IDE toolbar.

   c. Check the status bar at the bottom of the window; and wait for the text" Done Uploading".

   d. Upon successful upload, click the magnifying glass icon to open the serial monitor. Keep the device running for 3 minutes for system calibration.

   e. While waiting, position the thermometer and hygrometer directly proportional to each other 3 cm apart.

4. After 3 minutes of system calibration of the DHT22 sensor and microcontroller. Turn on the digital thermometer and hygrometer.

5. Once the DHT22 sensor, thermometer and hygrometer are all functioning properly, record all output readings. Enter the temperature readings from the DHT22 sensor and the thermometer in Table 3.1 under the row labeled "**Temperature Readings**". Record the humidity readings from the DHT22sensor and the hygrometer in Table 3.2 under the row labeled "**Humidity Readings**".

6. Compute the final average value of all the DHT22 sensor readings using:

$$x = \frac{\sum_{i=1}^{n} X_i}{n} \tag{3.1}$$

   where:

x = average value of DHT22 readings

X_i = temperature reading of DHT22 sensor

n = number of trials

7. Calculate the Mean Absolute Error (MAE), to determine the differences between the DHT22 sensor and thermometer using:

$$MAE = \frac{\sum_{i=1}^{n} |S_i - T_i|}{n} \tag{3.2}$$

where:

MAE = mean absolute error (average physical deviation in °C)

S_i = sensor reading for trial i

T_i = thermometer reading for trial i

n = number of trials

8. Calculate the Mean Percent Error (MPE), to determine the average error relative to the actual temperature:

$$\text{MPE} = \frac{1}{2} \sum_{i=1}^{n} \left(\frac{|S_i - T_i|}{T_i} \right) \times 100 \tag{3.3}$$

where:

MPE = average percentage of error across all trials, and

$\frac{|S_i - T_i|}{T_i}$ = relative error for one trial.

9. Compute for the overall accuracy (A) to determine the final system performance score using:

$$A = 100 - \text{MPE} \tag{3.4}$$

where:

A = the percentage of trials that match the reference standard (%).

10. Calculate the Coefficient of Variation (CV) to determine the precision and consistency of the DHT22 sensor using:

$$\text{CV} = \left(\frac{\sigma}{x}\right) \tag{3.5}$$

where:

σ = standard deviation of the data

x = mean value of the data

Record summary metrics in Table 3.3 in their respective columns and rows.

11. Repeat the process for 100 trials in the same workstation.

Table 3.1: DHT22 Sensor Accuracy at x°C

Temperature at x°C			
Trial	Thermometer Reading (℃)	DHT22 Sensor (℃)	%Accuracy
1	X		
...	X		
100	X		

Table 3.2: DHT22 Sensor Accuracy at x%

Humidity at x%			
Trial	Hygrometer Reading (%)	DHT22 Sensor (%)	%Accuracy
1	X		
...	X		
100	X		

3.2.2 Motion Sensor Accuracy Test

The objective of the experiment is to measure the detection accuracy and reliability of HCSR501 (PIR) Motion sensor detection. The materials are:

1. HCSR501 PIR motion sensor
2. ESP8266 Node MCU
3. 5m x 7m air-conditioned classroom
4. HCSR501_Calibration.ino
5. Protractor/Measuring tape
6. Wire
7. Microsoft Excel

The procedure is as follows:

1. Prepare all the materials needed. Mount HCSR501 motion sensor on the classroom wall/post at a height of 2.2 m, facing the center of the 7m x 5m room.
2. Open the Arduino IDE through a laptop, connect the ESP8266, then load the program for the motion sensor. Open the serial monitor to view the sensor output.
3. Power on the system and allow the sensor to stabilize for 60 seconds with no movement in the room.
4. Using a measuring tape, mark distances from the sensor at 1m interval, from 1m to 4m.

5. Establish angular reference lines relative to the centerline; 0° directly in front of the sensor, ±15°, ±30°, and ±45°
6. Mark these angles on the floor using tape or a protractor.
7. Establish the voltage-based decision rule for motion detection
8. If the PIR output voltage is $\geq 2.5\,\text{V}$, label the output as HIGH (motion detected).
9. If the PIR output voltage is $\approx 0\,\text{V}$, label the output as LOW (no motion detected).
10. Apply this decision rule consistently and record the labels (HIGH or LOW) in Table 3.3.
11. Position the test subject at 1 m distance and 0° angle.
12. The subject walks perpendicular to the sensor's field of view at normal walking speed.
13. Label the output as HIGH if the voltage is $\geq 2.5\,\text{V}$, or LOW if the voltage remains $\approx 0\,\text{V}$.
14. Record all the voltage readings, labels (HIGH/LOW) in the Table 3.3
15. Determine the detection rate per distance and angle in the formula as shown:

$$DR(d,\theta) = \frac{N_{\text{high}}}{100} \times 100 \tag{3.6}$$

where:

d = distance from the sensor

θ = angle from sensor centerline

N_{high} = number of trials with output voltage ≥ 2.5 V.

16. Record all the calculated detection rates per distance and angle in the Table 3.4

17. Repeat the test 100 times at the same distance and angle.

18. Repeat steps 11-17 for all marked angles (±15°, ±30°, ±45°) at the same distance.

19. Increase the distance to 2m, then repeat steps 11-18.

20. Continue testing for distances up to 4 m.

Table 3.3: V outputs 1 meter at 0°

1 meter at 0°		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.19: V PIR Motion detection results at 1 m distance

Distance (m)	Angle (°)	Total Trials	High Detection	Low Output	Detection Rate (%)
1	0	100			
1	15	100			
1	30	100			
1	45	100			

Table 3.4: V outputs 1 meter at $\pm 15^\circ$

1 meter at $\pm 15^\circ$		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.5: V outputs 1 meter at $\pm 30^{\circ}$

1 meter at $\pm 30^{\circ}$		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.6: V outputs 1 meter at $\pm 45^{\circ}$

1 meter at $\pm 45^{\circ}$		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.7: V outputs 2 meters at 0°

2 meters at 0°		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.8: V outputs 2 meters at $\pm 15^{\circ}$

2 meters at $\pm 15^{\circ}$		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.9: V outputs 2 meters at $\pm 30^{\circ}$

2 meters at $\pm 30^{\circ}$		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.10: V outputs 2 meters at $\pm 45^{\circ}$

2 meters at $\pm 45^{\circ}$		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.11: V outputs 3 meters at 0°

3 meters at 0°		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.12: V outputs 3 meters at $\pm 15^\circ$

3 meters at $\pm 15^\circ$		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.13: V outputs 3 meters at $\pm 30^\circ$

3 meters at $\pm 30^\circ$		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.14: V outputs 3 meters at $\pm 45^\circ$

3 meters at $\pm 45^\circ$		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.15: V outputs 4 meters at 0°

4 meters at 0°		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.16: V outputs 4 meters at $\pm 15^\circ$

4 meters at $\pm 15^\circ$		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.17: V outputs 4 meters at $\pm 30°$

4 meters at $\pm 30°$		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.18: V outputs 4 meters at $\pm 45°$

4 meters at $\pm 45°$		
Trial	Output Voltage (V)	Label (High/Low)
1		
n		

Table 3.20: V PIR Motion detection results at 2 m distance

Distance (m)	Angle (°)	Total Trials	High Detection	Low Output	Detection Rate (%)
2	0	100			
2	15	100			
2	30	100			
2	45	100			

Table 3.21: V PIR Motion detection results at 3 m distance

Distance (m)	Angle (°)	Total Trials	High Detection	Low Output	Detection Rate (%)
3	0	100			
3	15	100			
3	30	100			
3	45	100			

Table 3.22: V PIR Motion detection results at 4 m distance

Distance (m)	Angle (°)	Total Trials	High Detection	Low Output	Detection Rate (%)
4	0	100			
4	15	100			
4	30	100			
4	45	100			

3.2.3 Wireless Sensor Node Performance and Data Accuracy Test

The objective of the experiment is to measure the accuracy, stability, and reliability of the three sensor nodes.

The materials are:

1. 3 Sensor Nodes
2. Control Hub
3. 5m x 7m air-conditioned room
4. Sensor Nodes_Accuracy_test.ino
5. Wire
6. Microsoft Excel

The procedure is as follows:

1. Prepare all the materials needed for the experiment. Mount the three sensor nodes on the classroom wall at a height of 2.2 m, place each node in a fixed position (Left, Center, Right post) of the room.
2. For identification purposes, assign sensor node ID; Sensor node 1 (Left part), sensor node 2 (center), sensor node 3 (right part).

3. Open the Arduino IDE on the laptop and connect the ESP32 (control hub). Upload the receiver program and open the serial monitor to verify readiness for incoming data.

4. Power ON sensor node 1 and allow the system to stabilize for 60 seconds with no movement inside the classroom.

5. Define one transmission trial as the sensor node sending one complete data packet containing: Node ID, voltage output (V) for motion, Temperature (T) in ℃ and Humidity (H) in %.

6. Start the transmission process for sensor node 1

7. Allow the node to transmit exactly 100 consecutive packets every 10 seconds.

8. For each packet if the control hub receives the complete packet, mark the trial as successful.

9. For each packet is missing or corrupted, mark the trial as unsuccessful.

10. Record the data packet consists of voltage output, temperature, and humidity in the Table 3.5

11. Determine the transmission success rate for each sensor node as shown:

$$TSR = \frac{N_{scs}}{100} \times 100 \tag{3.7}$$

where:

TSR = transmission success rate (%),

N_{scs} = number of packets successfully received by the control hub.

12. Record the transmission success rate for each sensor node in Table 3.6

13. Repeat steps 4-12 for sensor node 2 and 3.

Table 3.23: Sensor Node 1 packet validation

Sensor Node 1				
Trial	PIR Voltage (V)	Temperature (°C)	Humidity (%)	Packet Status (Successful/Fail)
1				
...				
100				

Table 3.24: Sensor Node 2 packet Validation

Sensor Node 2				
Trial	PIR Voltage (V)	Temperature (°C)	Humidity (%)	Packet Status (Successful/Fail)
1				
...				
100				

Table 3.25: Sensor Node 3 Packet Validation

Sensor Node 3				
Trial	PIR Voltage (V)	Temperature (°C)	Humidity (%)	Packet Status (Successful/Fail)
1				
...				
100				

Table 3.26: Transmission Success Rate per Sensor Node

Sensor Node ID	Packets Sent	Packets Successfully Received	Failed Transmission	Transmission Success Rate (%)
Sensor Node 1	100			
Sensor Node 2	100			
Sensor Node 3	100			

3.2.4 Evaluation of AIRWise IR transmission Performance to Panasonic Window-type Inverter Air conditioner

The objective of the experiment is to measure the success rate that the aircon performs the expected action when the system sends command; and to measure success rate over many repeated trials.

The materials are:

1. AIRWise – automatic control system
2. Panasonic Window-type inverter air-conditioning unit
3. Wireless sensor nodes 1, 2, 3
4. 5m x 7m air-conditioned classroom
5. Microsoft Excel

The procedure is as follows:

1. Prepare all the materials and equipment to be used in this experiment.
2. Place the 3 sensor nodes, 2.0 meters above the floor level, in each area of the classroom, Left side, Middle, Right side
3. Place the AIRWise (Control Hub) at the back center of the room (X = 3.5 Meters, Y = 8 Meters, Height = 2 Meters)
4. Check all the nodes and the control hub connections and its positioning and make sure the room is empty
5. Check if the Aircon is OFF, Ensure the AC is in standby mode and can receive IR commands by using its original remote control

6. Start the test motion; five person walks within the detection range of the Sensor node 1, 3. Observe if the AIRWise detects motion (LED Blink), once motion is detected, AIRWise should automatically send the IR ON command to the aircon, then record the data.

7. Once the air conditioner is ON, AIRWise uses its logical control rules (programmed)

8. Adjust the environment manually or with a heat source/fan to change ambient readings, observe if AIRWise detects these changes and sends new IR commands accordingly.

9. Monitor the IR transmission by checking if the AC changes its mode (COOL, FAN, SWING) and target temperature automatically.

10. Record all the data and note the response time of each command to Table 3.7

11. Perform 100 trials per condition for consistent data.

Table 3.27: AIRWise evaluation test

Trial	Motion Detected	Temperature (°C)	Humidity (%)	Expected Mode	IR Sent?	AC Mode Changed?	Response Time

3.2.5 Voting Mechanism Accuracy Test

The objective of this experiment is to evaluate whether the voting mechanism can correctly adjust the air conditioner set temperature based on different vote combinations by comparing the expected and actual T_{snew} for each voting behavior test cases.

The materials are:

1. ESP32
2. IR Transmitter
3. Wireless Sensor Nodes 1, 2, 3
4. Microsoft Excel

The procedure is as follows:

1. Prepare all the materials: ESP32, IR transmitter, voting buttons (Warm, Slightly Warm, Slightly Cool, Cool), laptop with Arduino IDE, and 3 sensor nodes.
2. Set the air conditioner to Cool mode and stabilize at initial baseline temperature T_b of 22℃.
3. Connect the ESP32 to the laptop, open Arduino IDE, load the voting mechanism program, and open the serial monitor to view output.
4. For each test case in Table 3.7, set T_b to 22℃ or 24℃ as required and allow the air conditioner to stabilize.
5. Start the 5-minute voting session by pressing the first button, press buttons according to the selected vote pattern (W, SW, SC, C) from Table 3.7.
6. After completing the pattern, do not press any more buttons, wait until the full 5 minutes has elapsed and the program finishes the voting session.
7. When the session ends, record Actual T_{snew} from the serial monitor in Table 3.6 and confirm that $5 \leq N \leq 20$. discard invalid trials (N outside 5-20) before recording data.
8. Using T_b and ΔT from the serial monitor, determine the expected T_{snew} as shown below:

$$T_{\text{snew}} = T_b + \Delta T \tag{3.8}$$

where:

T_b = baseline temperature (°C),

ΔT = temperature change (°C).

9. Repeat steps 4 to 8 for 100 trials per test case.

10. For each case, decide the result: Pass if at least 95 out of 100 trials show the correct direction (Warm-dominant: $T_{\text{snew}} > T_b$; Cool-dominant: $T_{\text{snew}} < T_b$; Mixed: $T_{\text{snew}} \approx T_b$) and the actual T_{snew} is within $\pm 0.5\,^{\circ}$C of the expected T_{snew}; otherwise, Fail. Record the result in Table 3.7.

Table 3.28: Voting mechanism accuracy test

Case	Trial	T_b (°C)	Vote Combination	Expected T_{snew} (°C)	Actual T_{snew} (°C)	Result (Pass/Fail)
1	100	22	5, 0, 0, 0			
2	100	22	0, 0, 0, 5			
3	100	22	3, 3, 2, 2			
4	100	24	3, 3, 2, 2			
5	100	24	2, 0, 0, 0			

Chapter 4

RESULTS AND DISCUSSIONS

This chapter analyzes the data collected from the experiments. The results are supported by relevant tables, figures, and statistical analysis.

4.1 Temperature and Humidity Sensor Accuracy Test

In experiment 3.2.1, the DHT22 temperature and humidity sensor was performed by comparing its measured values against a calibrated digital thermometer and hygrometer used as the reference instrument. A series of repeated trials were conducted under stable indoor environmental conditions to minimize external thermal disturbances.

The reliability of the environmental monitoring system was evaluated through a series of calibration tests focusing on temperature and relative humidity. By comparing sensor outputs against high-precision reference instruments, the system's performance was quantified using four primary metrics: Mean Absolute Error (MAE), Mean Percent Error (MPE), Overall Accuracy (A), and Consistency (CV).

The linear curve was made by analyzing these data through the concept of linear regression [..] to show their relationship by scattering the data to get the linear equation. The linear curve is shown in Fig. 4.1. for temperature and Fig 4.2 for humidity. With this, the temperature and humidity equation is given by;

$$T_{DHT22} = (0.7457 \times T_{ref}) + 6.339 \tag{4.1}$$

where:

T_{DHT22} = Temperature values. and

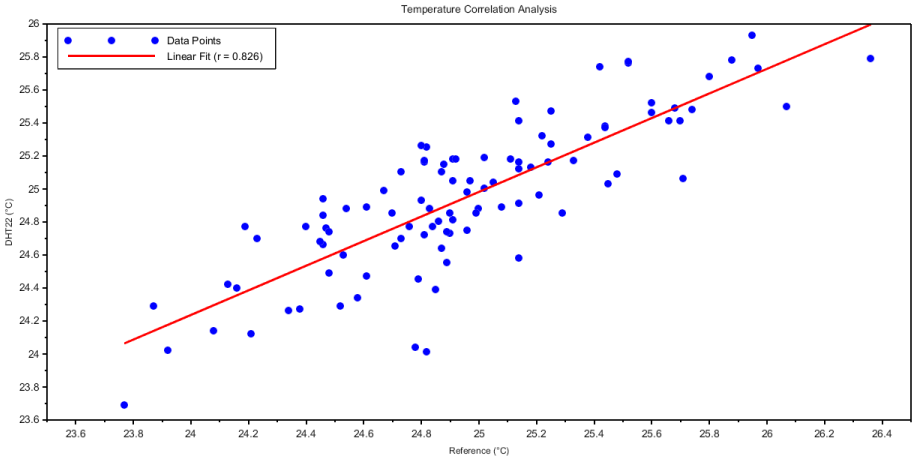


Figure 4.1: DHT22 (Temperature) Linear curve

T_{ref} = reference temperature

$$H_{DHT22} = (0.772 \times H_{ref}) + 12.4195 \tag{4.2}$$

where:

H_{DHT22} = humidity values and

H_{ref} = hygrometer reading.

This linear curve has a correlation of 0.826 and 0.878 which indicates a very strong linear relationship. This proves that as environment changes, DHT22 tracks the reference tool accurately

DHT22 demonstrated a very high level of precision in temperature measurement. With a Mean Absolute Error (MAE) of only 0.23℃ and an overall accuracy of 99.09%, the DHT22 sensor also performed well in measuring humidity, with an overall accuracy of 97.46%. Although the error in humidity is higher than that for temperature (MAE of 1.42% RH), this is well within the manufacturer's expected tolerance for DHT22 sensors. The sensor is highly reliable for temperature monitoring within the tested range (23℃–26℃), and it effectively captured fluctuations in environmental humidity.

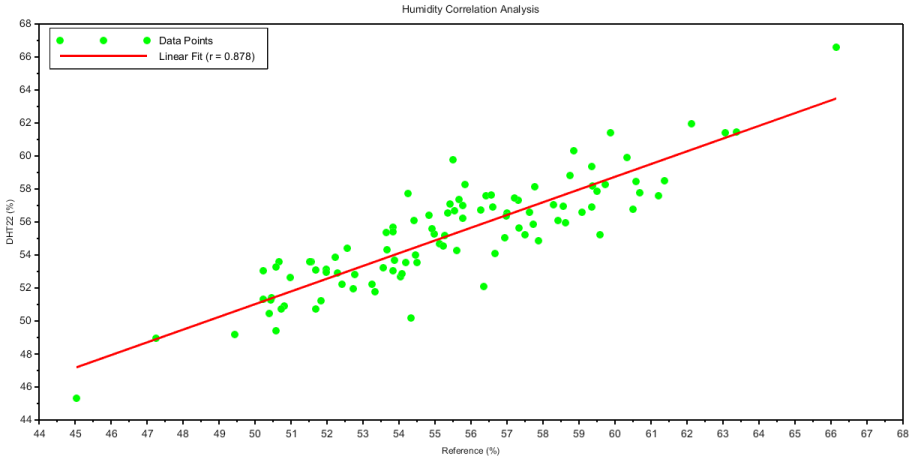


Figure 4.2: DHT22 (Humidity) Linear curve

The line graph was made by analyzing the relationship between the output data of the DHT22 sensor and the reference instruments. In Fig. **??**, The thermometer and DHT22 traces overlap tightly, with differences rarely exceeding to ±0.5℃, supporting the prior mean accuracy of 99%. Variability reflects natural room conditions, but the sensor tracks changes reliably without systematic drift. The same goes to hygrometer reading as shown in Fig. **??**.

4.2 Motion Sensor Accuracy Test

The experiment was conducted to evaluate the performance of the motion sensor under varying distances and angular positions. For each distance-angle configuration, two conditions were tested: with motion and without motion.

The results of the experiment indicate that the motion sensor maintained a consistently high level of detection performance across the tested distances and angles. As shown in Table 4.3,the highest detection rate recorded was 99% at 3 meters and an angle of 15°, demonstrating optimal sensor responsiveness under this configuration. In contrast, the lowest detection rate observed was 82% at a distance of 4 meters and an angle of 45°, which suggests a reduction in de-

Table 4.1: Motion Performance at 1-meter Distance

Distance (m)	Angle (°)	Total Trials	TP	FN	TN	FP	Accuracy	Sensitivity	Specificity
1	0	100	87	13	100	0	93.5%	87%	100%
1	15	100	92	8	100	0	96%	92%	100%
1	30	100	93	7	100	0	96.5%	93%	100%
1	45	100	85	15	100	0	92.5%	85%	100%

Table 4.2: Motion Performance at 2-meter Distance

Distance (m)	Angle (°)	Total Trials	TP	FN	TN	FP	Accuracy	Sensitivity	Specificity
2	0	100	91	9	100	0	95.5%	91%	100%
2	15	100	92	8	100	0	96%	92%	100%
2	30	100	91	9	100	0	95.5%	91%	100%
2	45	100	89	11	100	0	94.5%	89%	100%

tection capability as both distance and angular deviation increase. Despite this decrease at the farthest and widest configuration, the overall performance of the sensor remained strong, with detection rates consistently above 80% across all tested ranges. These findings confirm that the sensor operates reliably within the evaluated parameters while showing expected sensitivity variations at extended distances and wider angles.

4.3 Wireless Sensor Nodes Transmission Test

The experiment was performed by installing the sensor nodes at fixed positions inside a 5 m $\times$ 7 m air-conditioned room and monitoring both the transmitted and received data using separate laptops. Each transmission trial consisted of one complete data packet containing the Node ID, PIR voltage output, temperature, and humidity. A total of 100 consecutive packets were transmitted.

As shown in Table **??**, all 100 transmitted packets were successfully received

Table 4.3: Motion Performance at 3-meter Distance

Distance (m)	Angle (°)	Total Trials	TP	FN	TN	FP	Accuracy	Sensitivity	Specificity
3	0	100	94	6	100	0	97%	94%	100%
3	15	100	99	1	100	0	99.5%	99%	100%
3	30	100	89	11	100	0	94.5%	89%	100%
3	45	100	94	6	100	0	97%	94%	100%

Table 4.4: Motion Performance at 4-meter Distance

Distance (m)	Angle (°)	Total Trials	TP	FN	TN	FP	Accuracy	Sensitivity	Specificity
4	0	100	88	12	100	0	94%	88%	100%
4	15	100	91	9	100	0	95.5%	91%	100%
4	30	100	84	16	100	0	92%	84%	100%
4	45	100	82	18	100	0	91%	82%	100%

by the control hub, with zero failed transmissions recorded. This corresponds to a 100% transmission success rate. No packets were missing, incomplete, corrupted, or mismatched during the entire testing period. The consistent packet status shows that the communication link ensured accurate data transmission and stable synchronization between the sensor node and the control hub during the experiment.

Chapter 5

CONCLUSIONS AND RECOMMENDATIONS

Describe the organization or flow of the contents in this chapter before the start of the first section. What can your readers learn from this chapter? How did you organize your discussions sections? Section names should not be changed and subsections should not be added. Cross-references are also prohibited here. Your statements must be generalizations.

5.1 Conclusions

The first sentence should state what this thesis has accomplished. Make sure to connect or relate your objectives with your proposed method and the collected data. Statements should be based on the data, findings, and observations in the experiments. Conclusions that have no basis in previous chapters are **not acceptable**.

5.2 Recommendations

This section can include recommendations on how to improve your method. Improvement refers to technical aspects of the research and should not be about merely changing components or its aesthetic aspects. You can also discuss the problems or issues encountered or future works related to your research topic. Others identify the inherent limitations of their method then discuss how to overcome them.

NOTE: Before a recommendation, explain first the situation or its basis.

BIBLIOGRAPHY

Books

[1] R. C. Dorf and R. H. Bishop, *Modern Control Systems*, Thirteenth. Upper Saddle River, NJ: Pearson, Jan. 2017. [Online]. Available: https://www.pearson.com/en-us/subject-catalog/p/modern-control-systems/P200000003484?view=educator.

[2] K. J. Åström and T. Hägglund, *Advanced PID Control*, First. Research Triangle Park, NC: ISA - International Society of Automation, Aug. 2006. [Online]. Available: https://www.isa.org/store/advanced-pid-control/116187.

[3] J. S. Wilson, *Sensor Technology Handbook*, First. Oxford, UK: Elsevier, Jan. 2005. [Online]. Available: https://www.sciencedirect.com/book/9780750677295/sensor-technology-handbook.

[4] T. L. Floyd, D. M. Buchla, and S. Wetterling, *Electronic Devices (Conventional Current Version)*, Tenth. Hoboken, NJ: Pearson, Jan. 2017. [Online]. Available: https://www.pearson.com/en-us/subject-catalog/p/electronic-devices-conventional-current-version/P200000001047/9780134414447.

[5] J. W. Mitchell and J. E. Braun, "Evaporators and Condensers," in *Principles of Heating, Ventilation, and Air Conditioning in Buildings*, 1st, ME418 Course Material, Hoboken, NJ: John Wiley & Sons, Sep. 2024, ch. SM 7. [Online]. Available: https://www.purdue.edu/freeform/me418/wp-content/uploads/sites/30/2024/09/Chapter_SM_7_Evaporators_and_Condensers.pdf.

[6] J. G. Webster and H. Eren, *Measurement, Instrumentation, and Sensors Handbook: Spatial, Mechanical, Thermal, and Radiation Measurement* (Elec-

trical Engineering Handbook Series), 2nd. Boca Raton, FL: CRC Ppress, Jan. 2014, vol. 1. [Online]. Available: https://www.taylorfrancis.com/books/mono/10.1201/b15474/measurement-instrumentation-sensors-handbook-john-webster-halit-eren.

Published

[7] M. Benyamine, N. Fezzioui, M. A. Alkhafaji, N. Kaid, N. E. Benti, and Y. Menni, "Assessing indoor air quality and health risks of air conditioner usage in residential buildings under hot and arid climatic conditions: Insights from field studies and questionnaire analysis," *International Journal of Low-Carbon Technologies*, vol. 19, pp. 2397–2408, Oct. 2024.

[8] E. D. Cian, G. Falchetta, F. Pavanello, Y. Romitti, and I. S. Wing, "The impact of air conditioning on residential electricity consumption across world countries," *Journal of Environmental Economics and Management*, vol. 131, May 2025.

[9] G. Falchetta, E. D. Cian, F. Pavanello, and I. S. Wing, "Inequalities in global residential cooling energy use to 2050," *Nature Communications*, vol. 15, no. 7874, Sep. 2024.

[10] D. Aviv, A. Rysanek, N. Ma, and D. Vakalis, "Cooling People, Not Spaces: Surmounting the Risks of Air Conditioning Over-Reliance," Kleinman Energy Institute, University of Pennsylvania, Tech. Rep., Jan. 2025.

[11] J. Brooks, S. Kumar, S. Goyal, R. Subramany, and P. Barooah, "Energy-efficient Control of Under-actuated HVAC Zones in Commercial Buildings," *Energy and Buildings*, vol. 93, pp. 160–168, Apr. 2015. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S0378778815000687.

[12] S. Adepoju and S. David, “An Intelligent API Framework for Real-time Occupancy-Based HVAC Integration in Smart Building Management Systems,” *Journal of Knowledge Learning and Science Technology*, vol. 4, no. 1, pp. 61–70, Jan. 2025. [Online]. Available: https://jklst.org/index.php/home/article/view/v4.n1.007.

[13] J. Dong, C. J. Winstead, J. J. Nutaro, and P. T. V. Kuruganti, “Occupancy-Based HVAC Control with Short-Term Occupancy Prediction Algorithms for Energy-Efficient Buildings,” *Energies*, vol. 11, no. 9, p. 2427, Sep. 2018. [Online]. Available: https://www.ornl.gov/publication/occupancy-based-hvac-control-short-term-occupancy-prediction-algorithms-energy.

[14] J. G. Ordonez, C. Danielson, D. Limon, S. A. Bortoff, and S. Di Cairano, “Steady-State Analysis of HVAC Performance using Indoor Fans in Control Design,” in *IEEE Conference on Decision and Control (CDC)*, IEEE, Nice, France, Dec. 2019, pp. 2952–2957. [Online]. Available: https://ieeexplore.ieee.org/document/9029730.

[15] V. Intertechnology, *IR Receiver Modules for Remote Control Systems*, TSOP382 Datasheet, Vishay Semiconductors, Malvern, PA, Jan. 2023. [Online]. Available: https://www.vishay.com/docs/82491/tsop382.pdf.

[16] U. D. of Energy, “EnergyPlus Engineering Reference: Air System Fans,” Lawrence Berkeley National Laboratory, Berkeley, CA, Software Documentation Version 9.6, Sep. 2021. [Online]. Available: https://bigladdersoftware.com/epx/docs/9-6/engineering-reference/air-system-fans.html.

[17] J. W. Gauntner, J. N. B. Livingood, and P. Hrycak, “Survey of Literature on Flow Characteristics of a Single Turbulent Jet Impinging on a Flat Plate,” National Aeronautics and Space Administration, Lewis Research Center, Cleveland, Ohio, NASA Technical Note NASA TN D-5652, Feb. 1970. [On-

line]. Available: https://ntrs.nasa.gov/api/citations/19700009658/downloads/19700009658.pdf.

[18] *Temp Sensor for Air Conditioning*, Veteng Thinking Ltd., Dongguan, China, Jan. 2025. [Online]. Available: https://www.veteng.com/product/temp-sensor-for-air-conditioning.html.

Electronic Sources

[19] Hannah Ritchie, *Air conditioning causes around 3% of greenhouse gas emissions. how will this change in the future?* https://ourworldindata.org/air-conditioning-causes-around-greenhouse-gas-emissions-will-change-future, Jul. 2024.

[20] *Proportional, integral, derivative (PID) control*, https://www.eaton.com/in/en-us/products/controls-drives-automation-sensors/programamble-logic-controllers/hvac-control-for-plcs/temperature-control---hvac/pid-control---temperature-control---hvac.html, 2026.

[21] L. Suzhou CORESTAR Technology Co., *Air Conditioner Temperature Sensor: Classification, Function, and Resistance*, https://www.corestartech.com/blog/air-conditioner-temperature-sensor-classification-function-and-resistance/, Dec. 2025.

Appendix A

PRIOR ART SEARCH and ANALYSIS REPORT

Innovation and Technology Support Office
Cebu Institute of Technology - University
ITSO_IPPF002av01

ABSTRACT and KEYWORD TABLE

Name	
Program and Yr.	
Working Title	

Abstract
What does it do? Essential function of invention What is it made of? Physical structure of the invention What is the end result? Essential effect or basic product resulting from the invention What is it used for? Intended use for the invention

Keyword Table

SS01	SS02	SS03	SS04	SS05	SS06	SS07	SS08	SS08	SS10

CONFIDENTIAL

ABSTRACT and KEYWORD TABLE

Name	
Program and Yr.	
Working Title	

Abstract
What does it do? Essential function of invention What is it made of? Physical structure of the invention What is the end result? Essential effect or basic product resulting from the invention What is it used for? Intended use for the invention

Keyword Table

SS01	SS02	SS03	SS04	SS05	SS06	SS07	SS08	SS08	SS10

CONFIDENTIAL

IPC SEARCH REPORT

TITLE

SEARCH STRING	IPC	IPC DESCRIPTION	REMARKS
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard

CONFIDENTIAL

CERTIFICATION

SEARCH CONDUCTED BY	SEARCH REVIEWED BY
Signature and Date:	Signature and Date:
Complete Name:	Complete Name:
Department:	Department:
Institutional Research Code:	

PATENT SEARCH REPORT

TITLE	FILING DATE

ABSTRACT
What does it do? Essential function of invention What is it made of? Physical structure of the invention What is the end result? Essential effect or basic product resulting from the invention What is it used for? Intended use for the invention

CLASSIFICATION	KEYWORDS

CONFIDENTIAL

SEARCH STRINGS AND DATABASES

DATABASE	SEARCH STRING	NUMBER OF HITS

RELEVANT DOCUMENTS

CATEGORY	PATENT NO	TITLE/ABSTRACT	PUBLICATION DATE	RELEVANT TO CLAIM

* **Special Categories of Cited Documents:**

"A" document defining the general state of the art which is not considered to be of particular relevance

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

Note: Further references not indicated in this search report may be cited during substantive examination.

NON PATENT LITERATURE

CATEGORY	REFERENCE	RELEVANT TO CLAIM

PRIOR ART SUMMARY

PATENT NO. / REFERENCE TITLE	SIMILARITIES	DISSIMILARITIES	UNIQUE FEATURE

CONFIDENTIAL

CERTIFICATION

SEARCH CONDUCTED BY	SEARCH REVIEWED BY
Signature and Date:	Signature and Date:
Complete Name:	Complete Name:
Department:	Department:
Institutional Research Code:	

CONFIDENTIAL

Appendix A

PATENT DRAFT

Innovation and Technology Support Office
Cebu Institute of Technology - University
ITSO_IPPF005v01

SUBJECT MATTER ELIGIBILITY DETERMINATION FOR ICT AND CII

TITLE
ABSTRACT
What does it do? Essential function of invention What is it made of? Physical structure of the invention What is the end result? Essential effect or basic product resulting from the invention What is it used for? Intended use for the invention
PRIOR ART
STEP 1: Is the claim involving the field of ICT/CII?
STEP 2: Is the claim directed to a product or process within the meaning of the IP Code?
STEP 3: Does the claim involved an abstract idea?
STEP 4: Does the claim constitutes technical character wherein a technical solution to a problem is evident after weighing all the factors?

SEARCH CONDUCTED BY	
Signature and Date:	Signature and Date:
Complete Name:	Complete Name:
Department:	Department:

CONFIDENTIAL

INVENTION SPOTTING: INVENTIVE STEP SPOTTING

Claim	Your Invention:		CPA1: (number)	CPA2: (number)	CPA3: (number)
	(Title)		(Title)	(Title)	(Title)
TF1: TE:					
TF2: TE:					
TF3: TE:					
TF4: TE:					
TF5: TE:					
TP:					
TS:					

TF - Technical Feature TE - Technical Effect TP - Technical Problem TS - Technical Solution

A. 1. Find Novelty CPA (Y / N) 2. TFD - Technical Feature Difference (Y / N) 3. TED - TE Difference (Y / N) 4. Obvious? (Y / N)
5. Decision:

B. Modify TP? (Y / N):
Unexpected TE?:

CONFIDENTIAL

[insert figure, if more than one can fit in this page, merge this with the next]

Fig. 1

CONFIDENTIAL

APPLICANT'S NAME
Applicant

By:

YOUR NAME AND SIGNATURE
Agent

[insert figure, if more than one can fit in this page, merge this with the next]

Fig. 2

CONFIDENTIAL

APPLICANT'S NAME
Applicant

By:

YOUR NAME AND SIGNATURE
Agent

[insert figure, if more than one can fit in this page, merge this with the next]

Fig. 3

CONFIDENTIAL

APPLICANT'S NAME
Applicant

By:

YOUR NAME AND SIGNATURE
Agent

[insert figure, if more than one can fit in this page, merge this with the next]

Fig. 4

CONFIDENTIAL

APPLICANT'S NAME
Applicant

By:

YOUR NAME AND SIGNATURE
Agent

[insert figure, if more than one can fit in this page, merge this with the next]

Fig. 5

CONFIDENTIAL

APPLICANT'S NAME
Applicant

By:

YOUR NAME AND SIGNATURE
Agent

(TITLE)

(short and specific, if possible, not more than 10 words)

TECHNICAL FIELD OF THE INVENTION

(Statement of the field of the art to which the invention pertains, should be directed to the subject matter, starts with "BROAD" statement [broad field of technology] e.g. Computers, Electronics & Communications, and ends with "SPECIFIC" statement [specific field or goal] e.g. improving light sensor detection, usually one sentence only.

BACKGROUND OF THE INVENTION

(Cite Closest Prior Art Reference [CPAR] and state technical problems with [CPAR] that the invention wants to solve)

SUMMARY OF THE INVENTION

(State unexpected advantages of the invention over the CPAR)

BRIEF DESCRIPTION OF THE DRAWINGS

(Must show every feature of the invention, figures must be consecutively numbered, the elements must be designated by way of reference numerals corresponding to the elements in the description)

e.g. Figure 1 shows a top plan view of the invention.

CONFIDENTIAL

Figure 2 provides a block diagram of the invention.

DETAILED DESCRIPTION OF THE INVENTION

(State in detail the different parts or elements of the invention, how they interact, how the best embodiment can be manufactured or fabricated or replicated, and how the invention is used.)

(Broadening Clause)

The preferred embodiment of this invention is described in the above-mentioned detailed description. It is understood that those skilled in the art may conceive modifications and/or variations to the embodiment shown and described therein. Any such modifications or variations that fall within the purview of this description are intended to be included therein as well. Unless specifically noted, it is the intention of the inventors that the words and phrases in the specification and claims be given the ordinary and accustomed meanings to those of ordinary skill in the applicable art. The foregoing description of a preferred embodiment and best mode of the invention known to the applicant at the time of filing the application has been presented and is intended for the purposes of illustration and description. It is not intended to be exhaustive or to limit the present invention to the precise form disclosed, and many modifications and variations are possible in the light of the above teachings.

CONFIDENTIAL

CLAIMS (separate sheet)

1.
2.
3.
4.
5.

CONFIDENTIAL

(TITLE)

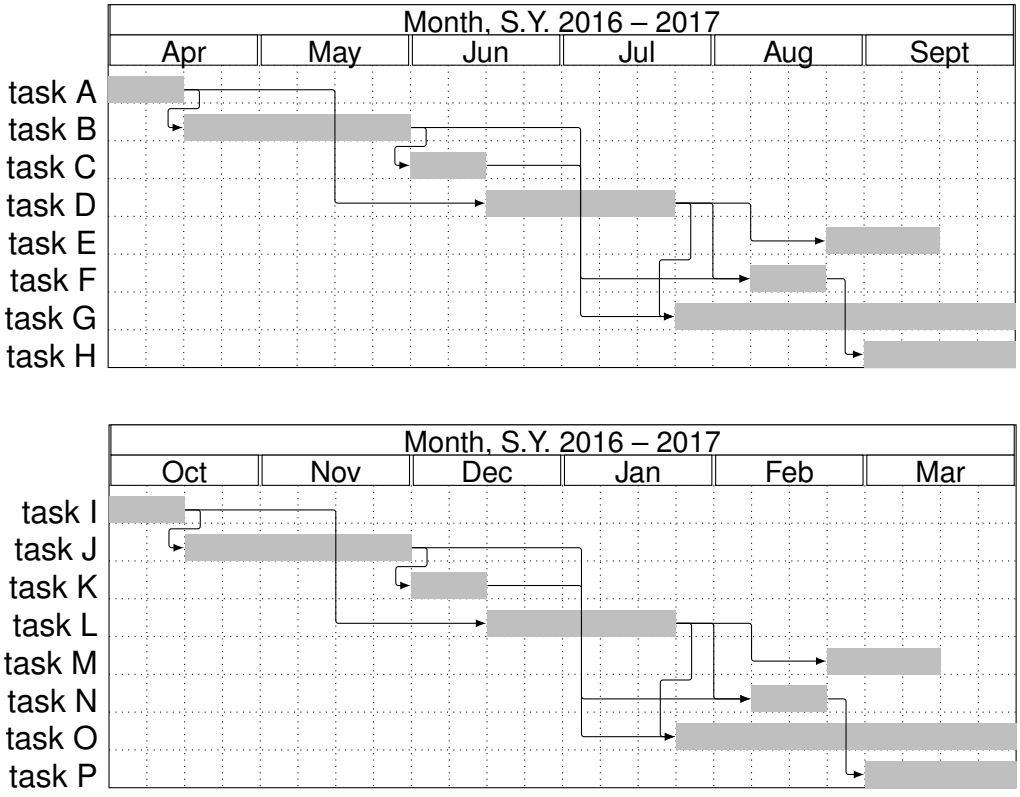
ABSTRACT OF THE DISCLOSURE (separate sheet)

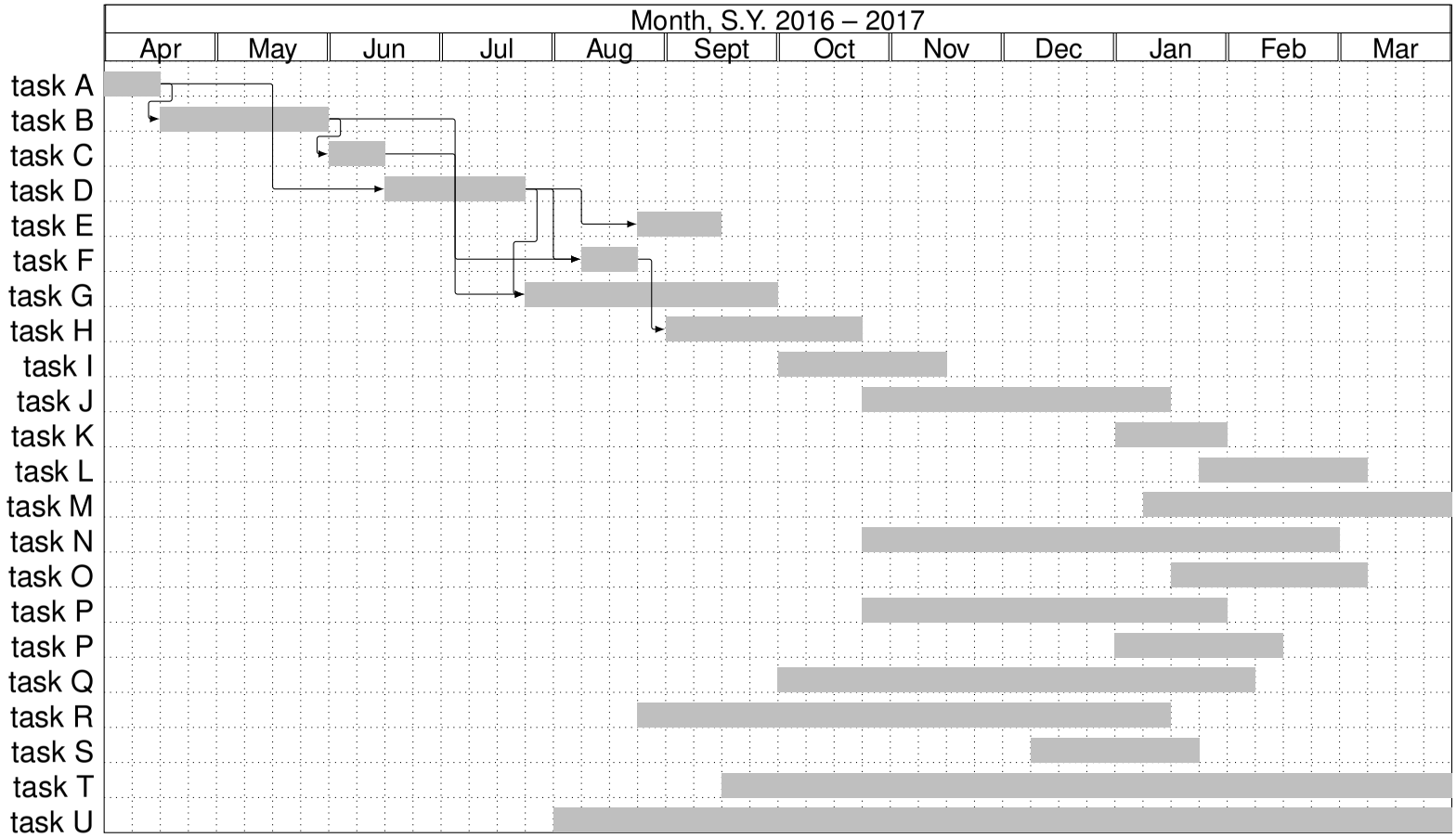
(State the technical problem that your invention solves and the elements/structures/parts of your invention, in 150 words)

CONFIDENTIAL

Appendix B

TIMETABLE OF ACTIVITIES




Month, S.Y. 2016 – 2017
Apr
May
Jun
Jul
Aug
Sept
Oct
Nov
Dec
Jan
Feb
Mar
task A
task B
task C
task D
task E
task F
task G
task H
task I
task J
task K
task L
task M
task N
task O
task P
task P
task Q
task R
task S
task T
task U

Task	Task Description	FamilyN1	FamilyN2	FamilyN3	FamilyN4
A	Specific activity				
B	Specific activity				
C	Specific activity				
D	Specific activity				
E	Specific activity				
F	Specific activity				
G	Specific activity				
H	Specific activity				
I	Specific activity				
J	Specific activity				
K	Specific activity				
L	Specific activity				
M	Specific activity				
N	Specific activity				
O	Specific activity				
P	Specific activity				
Q	Specific activity				
R	Specific activity				
S	Specific activity				
T	Specific activity				
U	Specific activity				

Appendix C

RESEARCH BUDGET

This is the budget of the thesis or project. During proposal, tabulate the possible expenses here. You can use a **long table** if your list of expenses spans several pages. In order to prevent your tables from floating, use the **tabular** environment only. List of expenses can be shorter by grouping them according to semester or a certain span of time. Item description and quantity should be left justified and centered, respectively. Amount should be right justified with two decimal places and comma for those that reach or exceed 1,000.00.

Appendix D

ADDITIONAL FLOWCHARTS

This appendix is for supplemental flowcharts only. Those that are for the main program or system should be inside Chapter 3. Figures here should also be cross-referenced or mentioned in other chapters.

Appendix E

PSEUDO-CODES

Insert pseudo-code of your algorithms here. Use the `algorithmic` environment. Below is an example:

Algorithm 1 Euclid's algorithm

```
1: procedure EUCLID(a, b)                    ▷ The g.c.d. of a and b
2:     r ← a mod b
3:     while r ≠ 0 do                        ▷ We have the answer if r is 0
4:         a ← b
5:         b ← r
6:         r ← a mod b
7:     end while
8:     return b                              ▷ The gcd is b
9: end procedure
```

If additional programs used in the experiment will be inserted, CHANGE the chapter title in the main tex file (line 173) to Experiment Codes. First, indicate the name of the program/code with file extension (ex. test1.sci) then enclose the commands with the verbatim environment.

Appendix F

ELECTRICAL CIRCUITS

Insert supplemental circuit diagrams here. Examples are power supply circuits (customized for the prototype) and driver circuits (not part of the main methodology). Insert the diagram as **figure** and cross-reference or mention these in other chapters.

Appendix G

SUPPLEMENTAL DATA

Insert your raw data here in tabular form (data from 100 trials). The tables MUST HAVE numbers and captions, i.e. use the **table** environment. If applicable, use long table with headers and footers. You can also cross-reference the tables here in your discussions in other chapters. Note that the data here should be summarized in a separate table or graph that is placed in Chapter 4.

Appendix H

PLAGIARISM CHECK REPORT

Final Doc

ORIGINALITY REPORT

9%	8%	4%	%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	en.wikipedia.org Internet Source	1%
2	ipindia.gov.in Internet Source	1%
3	patents.google.com Internet Source	1%
4	T.B. Asafa, T.M. Afonja, E.A. Olaniyan, H.O. Alade. "Development of a vacuum cleaner robot", Alexandria Engineering Journal, 2018 Publication	<1%
5	www.cnblogs.com Internet Source	<1%
6	repository.sustech.edu Internet Source	<1%
7	www.ebdigest.org Internet Source	<1%
8	www.worldbank.org Internet Source	<1%

9	www.makeuseof.com Internet Source	<1%
10	ijircce.com Internet Source	<1%
11	Vaibhav Palande, Adam Zaheer, Kiran George. "Fully Automated Hydroponic System for Indoor Plant Growth", Procedia Computer Science, 2018 Publication	<1%
12	N. Othman, M.I.A. Manan, Z. Othman, S.A.M. Al Junid. "Performance analysis of dual-axis solar tracking system", 2013 IEEE International Conference on Control System, Computing and Engineering, 2013 Publication	<1%
13	Rekha Chaudhary, Arun Kumar Singh. "Power system stabilization using an optimal hybrid control strategy with a PID controller", Journal of Electromagnetic Waves and Applications, 2018 Publication	<1%
14	ijesrt.com Internet Source	<1%
15	"Design and Development of an Artificial Intelligent System for Audio-Visual Cancer Breast Self-Examination", Journal of Advanced	<1%

	Computational Intelligence and Intelligent Informatics, 2016 Publication	
16	www.coursehero.com Internet Source	<1%
17	cps.kaist.ac.kr Internet Source	<1%
18	www.buydisplay.com Internet Source	<1%
19	www.ijert.org Internet Source	<1%
20	www.studymode.com Internet Source	<1%
21	infocenter.nordicsemi.com Internet Source	<1%
22	vdocuments.site Internet Source	<1%
23	Wen-long Cheng, Jia-liang Song, Yi Liu, Shuai Yuan, Wan-fan Wu, Zhi-ming Xu. "Theoretical and experimental studies on thermal control by using a novel PTC material with room temperature Curie point", International Journal of Heat and Mass Transfer, 2014 Publication	<1%
24	t-r-a-v-e-l-new-orleans.com	

	Internet Source	<1%
25	Abu-Zreig, M.M.. "The auto-regulative capability of pitcher irrigation system", Agricultural Water Management, 20061016 Publication	<1%
26	fivedots.coe.psu.ac.th Internet Source	<1%
27	"Intelligent Robotics Systems: Inspiring the NEXT", Springer Nature America, Inc, 2013 Publication	<1%
28	Ahmed Al-Timimy, Gaurang Vakil, Michele Degano, Paolo Giangrande, Chris Gerada, Michael Galea. "Considerations on the Effects that Core Material Machining has on an Electrical Machine's Performance", IEEE Transactions on Energy Conversion, 2018 Publication	<1%
29	www.omicsonline.org Internet Source	<1%
30	cdm16322.contentdm.oclc.org Internet Source	<1%
31	treeprotectionsupply.com Internet Source	<1%
32	repository.up.ac.za Internet Source	

		<1%
33	www.nxp.com Internet Source	<1%
34	www.freepatentsonline.com Internet Source	<1%
35	Sean Tagle, Rica Pena, Frances Oblea, Hans Benoza, Nadine Ledesma, Jeremias Gonzaga, Laurence A. Gan Lim. "Development of an Automated Data Acquisition System for Hydroponic Farming", 2018 IEEE 10th International Conference on Humanoid, Nanotechnology, Information Technology,Communication and Control, Environment and Management (HNICEM), 2018 Publication	<1%
36	psasir.upm.edu.my Internet Source	<1%
37	Devi, M. Jyostna, Thomas R Sinclair, Mukesh Jain, and Maria Gallo. "Leaf aquaporin transcript abundance in peanut genotypes diverging in expression of the limited-transpiration trait when subjected to differing vapor pressure deficits and aquaporin inhibitors", Physiologia Plantarum, 2015. Publication	<1%

38	tigerprints.clemson.edu Internet Source	<1%
39	H.C.H. Ho, W.L. Cheung, I. Gibson. "Effects of graphite powder on the laser sintering behaviour of polycarbonate", Rapid Prototyping Journal, 2002 Publication	<1%
40	scholarscompass.vcu.edu Internet Source	<1%
41	eprints.uthm.edu.my Internet Source	<1%
42	arxiv.org Internet Source	<1%
43	archive.org Internet Source	<1%
44	mylistahan.com Internet Source	<1%
45	spiral.imperial.ac.uk Internet Source	<1%
46	Deny Prasetia Taruma Wardana, Fatwa Ramdani, Fajar Pradana. "Real-Time Landslide Monitoring System Based on 3D WebGIS and Embedded System", 2018 4th International Symposium on Geoinformatics (ISyG), 2018 Publication	<1%

47	www.dtic.mil Internet Source	<1%
48	"Proceedings of the 10th National Technical Seminar on Underwater System Technology 2018", Springer Nature, 2019 Publication	<1%
49	Mitar Simic. "Design and development of air temperature and relative humidity monitoring system with AVR processor based web server", 2014 International Conference and Exposition on Electrical and Power Engineering (EPE), 2014 Publication	<1%
50	"Advances in Signal Processing and Intelligent Recognition Systems", Springer Nature America, Inc, 2016 Publication	<1%
51	L. Elvira, L. Sampedro, F. Montero de Espinosa, J. Matesanz, Y. Gómez-Ullate, P. Resa, F.J. Echevarría, J.R. Iglesias. "Eight-channel ultrasonic device for non-invasive quality evaluation in packed milk", Ultrasonics, 2006 Publication	<1%
52	Rahman K. Fahmidur, Hossain M. A. Munaim, Sadiq M. Tanvir, Ahmad S. Sayem. "Internet	<1%

controlled robot: A simple approach", 2016 International Conference on Electrical, Electronics, and Optimization Techniques (ICEEOT), 2016
Publication

Exclude quotes	Off	Exclude matches	Off
Exclude bibliography	On		

Appendix I

ETHICS CLEARANCE

CEBU INSTITUTE OF TECHNOLOGY – UNIVERSITY
Institutional Ethics Review Committee
ethics@cit.edu

Month DD, YYYY

COMPLETE NAME OF PROPONENT/GROUP LEADER
Designation
Institution
Address

RE: **Title of project/study**
IERC Code: **Code**

Subject: Research Ethical Clearance

Dear **Name of Proponent/Group Leader**:

This is to acknowledge receipt of your request and the following supporting documents dated Month DD, YYYY.

1. Submitted document 1
2. Submitted document 2
3. Submitted document 3
4. Submitted document 4
5. Submitted document 5

The above documents underwent FULL/EXPEDITED review (for FULL:) during the meeting on Month DD, YYYY at Venue of Meeting.

The study protocol is hereby granted APPROVAL FOR IMPLEMENTATION by CIT-U IERC. This ethical clearance is valid for one year until Month DD, YYYY.

While the study is in progress, the Proponent/s is/are requested to submit the following documents:

1. Progress report using form IERC-PF-19: Application for Continuing Review every six months from date of approval including the following: (*NOTE: In view of the active ethical clearance, this report is mandatory even if the study has not started or is still awaiting release of funds.*)

    a. Date covered by the report
    b. Protocol summary and status report on the progress of the research

c. Number of participants accrued
d. Withdrawal or termination of participants
e. Complaints on the research since the last CIT-U IERC review
f. Any relevant multi-center research reports
g. Any relevant information especially about risks associated with the research
h. A copy of the informed consent document

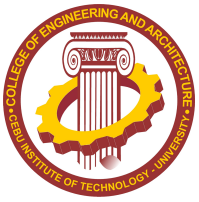
2. Any changes in the protocol, especially those that may adversely affffect the safety of the participants during the conduct of the study, including changes in personnel, must be submitted or reported using form IERC-PF-15: Application for Ethics Review of Amendments.
3. Revisions in the informed consent form using form IERC-PF-15: Application for Ethics Review of Amendments.
4. Reports of adverse events including from other study sites using the form IERC-PF-18: Reportable Negative Event Report.
5. Notice of early termination of the study and reasons for such using form ERC-PF-21: Early Termination Report.
6. Any event which may have ethical significance.
7. Any information which is needed by the CIT-U IERC to do ongoing review
8. Notice of time of completion of the study using IERC-PF-22: Final Report.
9. Application for renewal of ethical clearance 90 days before the expiration date of this approval through submission of form ERC-PF-19: Application for Continuing Review.

Very truly yours,

CHAIR's COMPLETE NAME
Chair

Appendix J

CURRICULUM VITAE



CONTACT INFORMATION

Name: Given Name MI. Family Name

Address: Number and Street, City or Province, Philippines, Zip Code

Telephone Number: 063 - area code - number

Cell Phone Number: 09numbers

Email: address@site.dmn

PERSONAL INFORMATION

Date of Birth: Month DD, YYYY
Place of Birth: City, Country
Citizenship: Filipino

EDUCATIONAL BACKGROUND

College
Cebu Institute of Technology University
N. Bacalso Avenue, Cebu City, Philippines
Bachelor of Science in Electronics Engineering
March 2013

High School
Name of High School attended
Address
Month YYYY

WORK EXPERIENCE

Name of Company (latest)
Address
Position
Responsibilities
Inclusive Dates

Name of Company
Address
Position
Responsibilities
Inclusive Dates

PROFESSIONAL QUALIFICATIONS

Certifications
certification 1 (latest)
certification 2

Computer Skills
skill 1
skill 2

HONORS AND AWARDS

Name of Award 1 (latest)
Institution
Month, YYYY

Name of Award 2
Institution
Month, YYYY

PROFESSIONAL AFFILIATION

Name of Organization 1 (latest)
Position
Inclusive Years

Name of Organization 2
Position
Inclusive Years

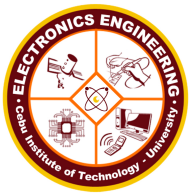
SKILLS AND INTERESTS

Skills: skill 1, skill 2, skill 3

Software: software 1, software 2, software 3, software 4, software 5

Community Extension: extension 1, extension 2

Languages: Language 1 - Native, Language 2 - Fluent, Language 3 - Advanced or Intermediate or Beginner



CONTACT INFORMATION

Name: Given Name MI. Family Name

Address: Number and Street, City or Province, Philippines, Zip Code

Telephone Number: 063 - area code - number

Cell Phone Number: 09numbers

Email: address@site.dmn

PERSONAL INFORMATION

Date of Birth: Month DD, YYYY
Place of Birth: City, Country
Citizenship: Filipino

EDUCATIONAL BACKGROUND

College
Cebu Institute of Technology University
N. Bacalso Avenue, Cebu City, Philippines
Bachelor of Science in Electronics Engineering
March 2013

High School
Name of High School attended
Address
Month YYYY

WORK EXPERIENCE

Name of Company (latest)
Address
Position
Responsibilities
Inclusive Dates

Name of Company
Address
Position
Responsibilities
Inclusive Dates

PROFESSIONAL QUALIFICATIONS

Certifications
certification 1 (latest)
certification 2

Computer Skills
skill 1
skill 2

HONORS AND AWARDS

Name of Award 1 (latest)
Institution
Month, YYYY

Name of Award 2
Institution
Month, YYYY

PROFESSIONAL AFFILIATION

Name of Organization 1 (latest)
Position
Inclusive Years

Name of Organization 2
Position
Inclusive Years

SKILLS AND INTERESTS

Skills: skill 1, skill 2, skill 3

Software: software 1, software 2, software 3, software 4, software 5

Community Extension: extension 1, extension 2

Languages: Language 1 - Native, Language 2 - Fluent, Language 3 - Advanced or Intermediate or Beginner